



Document Image Quality Assessment: A Survey

Alaei, Alireza; Bui, Vinh; Doermann, David; Pal, Umapada

<https://researchportal.scu.edu.au/esploro/outputs/journalArticle/Document-Image-Quality-Assessment-A-Survey/991013125928502368/filesAndLinks?index=0>

Alaei, A., Bui, V., Doermann, D., & Pal, U. (2024). Document Image Quality Assessment: A Survey. ACM Computing Surveys, 56(2), Article 29. <https://doi.org/10.1145/3606692>
Document Version: Accepted

Published Version: <https://doi.org/10.1145/3606692>

Southern Cross University Cross Connect: <https://researchportal.scu.edu.au/esploro/crossconnect@scu.edu.au>

Free to Read

Research:Open

© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.

Downloaded On 2026/09/30 05:46:43 +1000



Document Image Quality Assessment: A Survey

Alireza Alaei

Faculty of Science and Engineering, Southern Cross University, Australia, ali.alaei@scu.edu.au

Vinh Bui

Faculty of Science and Engineering, Southern Cross University, Australia, vinh.bui@scu.edu.au

David Doermann

University at Buffalo, Buffalo, NY, The USA, doermann@buffalo.edu

Umapada Pal

Indian Statistical Institute, Kolkata, India, umapada@isical.ac.in

The rapid emergence of new portable capturing technologies has significantly increased the number and diversity of document images acquired for business and personal applications. The performance of document image processing systems and applications depends directly on the quality of the document images captured. Therefore, estimating the document's image quality is an essential step in the early stages of the document analysis pipeline. This paper surveys research on Document Image Quality Assessment (DIQA). We first provide a detailed analysis of both subjective and objective DIQA methods. Subjective methods, including ratings and pair-wise comparison-based approaches, are based on human opinions. Objective methods are based on quantitative measurements, including document modeling and human perception-based methods. Second, we summarize the types and sources of document degradations and techniques used to model degradations. In addition, we thoroughly review two standard measures to characterize document image quality: Optical Character Recognition (OCR)-based and objective human perception-based. Finally, we outline open challenges regarding developing DIQA methods and provide insightful discussion and future research directions for this problem. This survey will become an essential resource for the document analysis research community and serve as a basis for future research.

CCS CONCEPTS: • **General and reference** → **Surveys and overviews** • **Computing methodologies** → **Artificial intelligence** • **Applied computing** → **Document management and text processing**

Additional Keywords and Phrases: Document Image Quality, Document Image Readability, Image Quality Assessment

1 INTRODUCTION

A traditional physical document is a printed or handwritten manuscript commonly used to transfer/convey information over time and space [1]. Technological advances have made physical documents colorful with more complex structures containing tables, figures, pictures, logos, barcodes, etc. An example of such documents is provided in Figure 1. The quality of a document is always an important issue for both senders and receivers. Documents of inferior quality may be harder to read, while high-quality documents typically allow a more effortless transfer of information. Document quality is often subjective, especially concerning handwriting.

Nevertheless, human perception might describe the overall visual appearance and compare it to an idealized version of a similar document. While we review document quality for information transfer, documents can also be evaluated based on their artistic beauty. Artistic beauty is considered beyond the scope of this survey.

As we refer to in this paper, a document image is a digital representation of a physical document. An optical imaging device, such as a scanner, smartphone camera, or digital still camera, often acquires it. Similar to the quality of the original document, the quality of its digital representation is also subjective. ISO TC 42 committee has defined *image quality* as the: "impression of the overall merit or excellence of an image, as perceived by an observer..." [2]. This definition is based on human perception's psychophysical process and the standard for estimating image quality using psychophysical methods (ISO 20462) [3]. This definition may not be applied directly to document images, and to the best of our knowledge, there is no specific ISO definition for document image quality. In this paper, we refer to the term quality as the readability of the document image. Document

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.

0360-0300/2023/1-ART1 \$15.00

<http://dx.doi.org/10.1145/3606692>

images are commonly described as having background and foreground information, where the foreground contains the content one is interested in analyzing. The Human Visual System (HVS) can typically differentiate these two, assuming some regions in a document image have greater relevance for human observers [4]. Document image readability, however, can be measured from a human or machine perspective. Human readability is estimated based on an individual's ability to acquire the content correctly, whereas machine readability is assessed using Optical Character Recognition (OCR) accuracy or another empirical measure. In human and machine readability, poor-quality document images often result in low document image readability. The terms document image "readability" and "quality" have, therefore, been used in the literature interchangeably [5]–[7].

Southern Cross University

Contact Form

Please fill in this form and send it to the Student Service Desk. Student Service Desk will contact you and provide more information about the University and can answer your questions.

Name

First Last

Address

Address Line 1

Address Line 2

City State / Province / Region

Postal / Zip Code Country

Phone **Email**

Contact Us
Email: enquiry@scu.edu.au Phone: 1800 005 887



Figure 1: An example of physical documents.

New digitization technologies, such as smartphone cameras, tablets, portable scanners, digital cameras, and other gadgets, have empowered individuals to generate more data from documents in images and videos. In addition, public and private organizations and businesses are equipping themselves with new technologies that allow users/clients to access their systems, perform data entry and upload digitized documents as a part of an attempt to move toward a paperless environment. As a result, a massive number of images, including document images, are generated every day. As the performances of almost all document image processing applications, including document image classification, segmentation, recognition, retrieval, and even restoration, are directly dependent on the quality of the initial document images, acquiring or maintaining a quality source image is vital to the success or failure of those systems or applications. For that reason, quality estimation is essential at various stages of acquisition. For example, in an automatic banking system, if a cheque of inferior quality is submitted to the system to claim the amount, the system should quickly send a warning to the sender indicating that the quality of the submitted cheque image is insufficient for processing. Similarly, if the quality is inadequate for automated processing in a postal application, the address block can be routed to a human for manual interpretation. As a result, understanding and estimating document image quality followed by image enhancement are essential to mitigate the negative impact of poorly quality document images and degradations on different applications.

Image quality is generally considered subjective but is often measured relative to a Human Opinion Score (HOS). Given the considerable volume of document images generated, human-involved document image quality estimation could be more realistic and practical. An alternative is to train a classifier to mimic human scores. One consideration, however, is that in a processing pipeline, the goal is not human readability but the accuracy

of a downstream process. The relative quality we wish to assess may be task-dependent. For example, if we need to perform OCR, we may want to measure OCR accuracy as a metric for document image quality. We may then ask, “Do we need human opinion scores?” There are undoubtedly several considerations in this case. First, OCR engines do not always perform well on handwritten documents, historical documents with old styles and fonts, or low-resource languages. Second, OCR accuracies do not always correlate with visual quality. Third, OCR may only represent the quality of the text content of the image and not other content, such as graphics. Finally, we often may not know what assumptions the OCR makes about the content. An OCR may also use comprehensive natural language processing (NLP) models, such as Generative Pre-trained Transformers (GPT) [8] and Bidirectional Encoder Representations from Transformers (BERT) [9], to obtain more accurate OCR output. In addition, an OCR system may do very well with one document type but fail miserably with others [10]–[13].

A significant number of algorithmic Document Image Quality Assessment (DIQA) methods have been proposed in the past to address these problems [14], [15]. Ye and Doermann [14] briefly presented the latest review of DIQA methods in the literature. There are several survey papers in the literature for IQA and face IQA (FIQA), for example, [16], [17], which provide researchers with a detailed overview of advances in these research domains. Considering the DIQA as an active and growing area of research in document image analysis, it is essential to provide researchers with an update on technologies currently available for DIQA and highlight advances and progress achieved in the past. It is also important to point out the directions for future research in this domain. Therefore, this survey presents state-of-the-art, subjective DIQA methods and their characteristics. Practices proposed for objective DIQA in the literature, including modeling-based approach and human perception-based techniques, are reviewed. Document modeling and understanding-based methods, types and sources of document degradation followed by document image degradation modeling are thoroughly discussed. OCR-based methods for DIQA are also detailed. Furthermore, human perception-based (objective) DIQA measures and metrics are reviewed to characterize document image quality. Finally, a comprehensive discussion followed by the future of research in this domain is outlined to provide researchers with insights and new research directions.

The remainder of this survey is organized as follows. Section 2 provides a general overview of document image quality assessment. Sections 3 and 4 present a comprehensive review of DIQA methods in the literature. Section 5 discusses evaluation metrics, benchmarks, and datasets used for experimental and comparative analysis. Section 6 provides a broad comparison of state-of-the-art methods. Section 7 contributes to the discussion and remarks on the methods proposed in the literature and future research directions.

2 AN OVERVIEW OF DIQA

2.1 Document Image Analysis and the Role of DIQA

Figure 2 demonstrates the document digitization process and possible degradations at different stages of a document image analysis pipeline and the role of DIQA in this pipeline. In the document image analysis pipeline, different types of degradations may occur during a document life cycle that may reduce the information or visual quality of the document concerning its original [12], [14].

First, degradations may occur during **document creation** due to a low-quality medium (e.g., paper translucency or texture), devices used to create the document (e.g., inconsistent ink flow or poorly maintained equipment), or the production process (e.g., inadequate or heavy printing, handwriting imperfections, or noise in electronic components). Second, the physical documents' **existence over time** may subject them to external or physical degradations. These may include human and environmental impacts, such as paper aging, stains, torn-off regions, reader marking and annotations, carbon copy effect, scratches, and cracks [18], [19]. Third, document **digitization** may introduce degradations caused by analog-to-digital conversion, which loosely converts physical documents into digital representations. In addition, distortions such as shade along the “spine” of the book and warping in the shaded area may be introduced due to device flaws and/or vibration, improper paper positioning (skew), sensor sensitivity variations, and the thick bound volume of the books when using

scanners, mobile phones, and cameras during the digitization process [20]–[23]. Fourth, digital-born and digitized documents may undergo further **image processing**, including encoding, compression, and transformation, to make them suitable for analysis, storage, or transmission over networks. For example, as two document image processing methods, image binarization and lossy compression may introduce binarization and JPEG or JPEG2k compression artifacts to document images. Finally, the quality of **transmission** networks may affect the quality of document images on the receiver side. It is worth noting that a document may go through multiple stages of printing, digitization, compression, and transmission, causing repeated and combined degradations in unexpected ways. Due to document image degradations, the DIQA (human perception or modeling based) is crucial in the processing pipeline to decide whether a document image needs manual processing or quality enhancement operations.

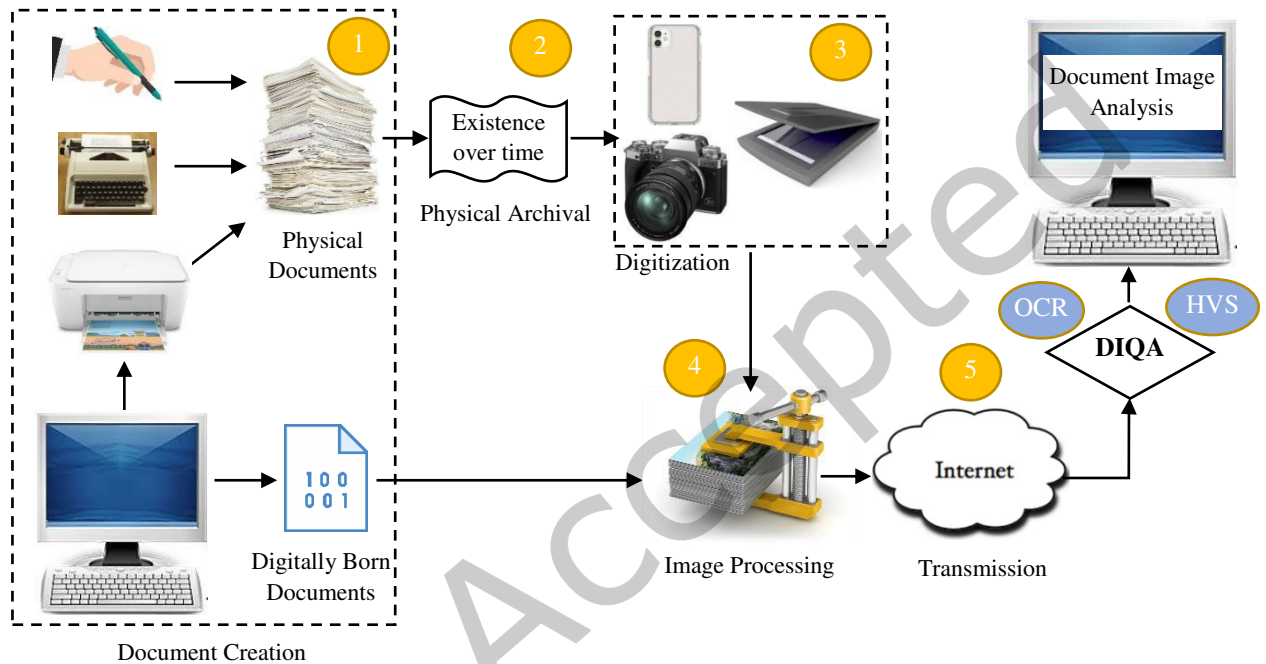


Figure 2: DIQ is affected by the document life cycle.

2.2 DIQA problems and development

The problem of DIQA has initially been addressed by developing several document degradation models and trying to enhance the quality of document images in the early 90s (see Figure 3(a)). As shown in Figure 3(a), document image degradation research's highest growth was from 2001-2005. The research then slowly shifted to the design and development of DIQ metrics, reaching its peak during 2016-2022, as shown in Figure 3(a). Figure 3(b) further demonstrates the growth and shift in technology from OCR based to perception-based metrics. It is also evident that OCR accuracy and OCR-based metrics were the dominant DIQA metrics during 1991-2000, whereas perception-based metrics became the trend in recent years, during 2016-2022. The frequency of keywords used in the literature on DIQA also demonstrates that document image quality and related keywords followed by document degradation modeling have frequently been used in publications in this domain (Figure 3(c)).

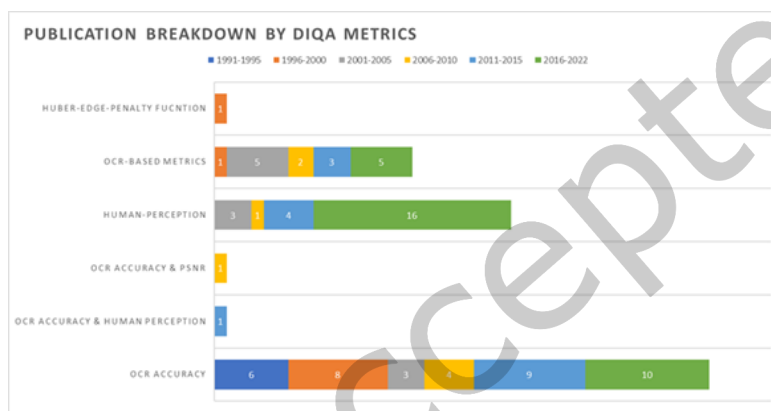
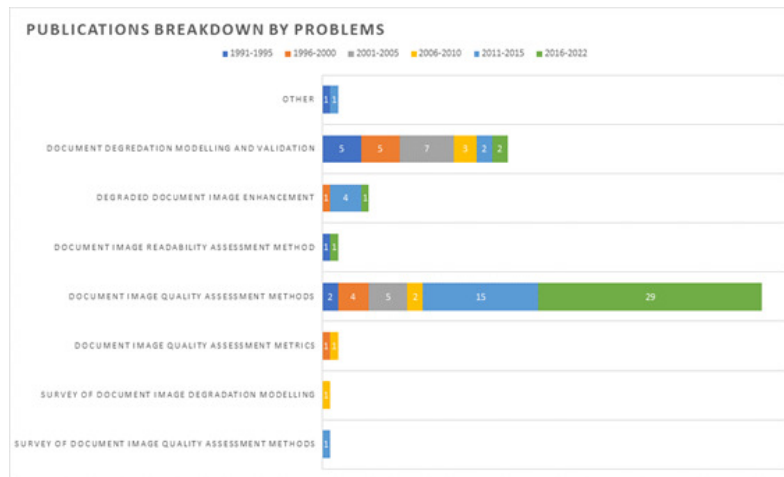


Figure 3: The development and growth of DIQA between 1990 and 2022, (a) publications in different years, (b) publications by metrics, and (c) keywords cloud of papers reviewed in this research work.

2.3 DIQA Methods and Metrics

The earliest attempts to deal with the problem of IQA and DIQA in the literature are based on the Mean Squared Error (MSE) computed by averaging the squared intensity differences of distorted and reference image pixels and the Peak Signal-to-Noise Ratio (PSNR) [24]–[27]. These two metrics are simple to compute, but they cannot characterize visual image quality [25], [27] and document image quality [26], [28]–[33]. In addition, to

get an idea of the performance of the PSNR and MSE in the DIQA domain, a sample text document and two degraded versions derived from the given document, along with their corresponding PSNR, MSE, and OCR accuracies, are provided in Figure 4. The results shown in Figure 4 indicate that OCR has produced an identical OCR accuracy for all three images, while PSNR and MSE are entirely different. This indicates that these metrics are not effective for DIQA.

Therefore, many sophisticated methods have been proposed in the literature to provide more accurate document image quality measurements. The hierarchical classification of DIQA methods in the literature is shown in Figure 5. As shown in Figure 5, DIQA methods can be grouped into two main categories, subjective and objective. Subjective approaches in the literature are classified into pair-wise and ratings-based methods.

Mr. Darling used to boast to Wendy that her mother not only loved him but respected him. He was one of those deep ones who know about stocks and shares. Of course no one really knows, but he quite seemed to know, and he often said stocks were up and shares were down in a way that would have made any woman respect him.

(a) An original text image from the ULR-textSISR-2013a dataset [34] and its OCR accuracy (OCR=99.7)

Mr. Darling used to boast to Wendy that her mother not only loved him but respected him. He was one of those deep ones who know about stocks and shares. Of course no one really knows, but he quite seemed to know, and he often said stocks were up and shares were down in a way that would have made any woman respect him.

(b) A degraded (2 pixels translated) version of (a), PSNR = 36.25 dB, MSE = 15.42 and OCR = 99.7

Mr. Darling used to boast to Wendy that her mother not only loved him but respected him. He was one of those deep ones who know about stocks and shares. Of course no one really knows, but he quite seemed to know, and he often said stocks were up and shares were down in a way that would have made any woman respect him.

(c) A degraded version of (a) by adding salt and pepper noise, PSNR = 45.68 dB, MSE = 1.76 and OCR=99.7

Figure 4: A sample text (original) from [Super] and its degradation versions

Objective DIQA can further be divided into document understanding (modeling-based) and human perception-based approaches. Document understanding and modeling-based approach cover methods that try to understand and model image degradations and incorporate the results of an OCR system along with document image characteristics to predict the qualities of document images. The perception-based approaches include full-reference (FR) DIQA and no-reference (NR) DIQA. General frameworks for FR- and NR-DIQA are provided in Figures 6(a) and 6(b). To the best of our knowledge, there is no RR-DIQA method in the literature. For the sake of completeness of the presentation, a general framework for RR-DIQA is also presented in Figure 6(c). As shown in Figure 6a, the FR-DIQA approach considers both original and distorted document images to estimate the quality of the distorted image. At the same time, the NR-DIQA approach considers only a single image as input to the system to assess its quality (Figure 6b). The RR-DIQA approach utilizes a loss-less channel to transfer the extracted features from the input image and a noisy channel to transfer the input image (Figure 6c). An RR-DIQA, similar to FR-DIQA, finally estimates the quality of the transmitted image on the receiver side. In Figure 6c, the term noisy channel is referred to any component in the processing pipeline where degradations may occur.

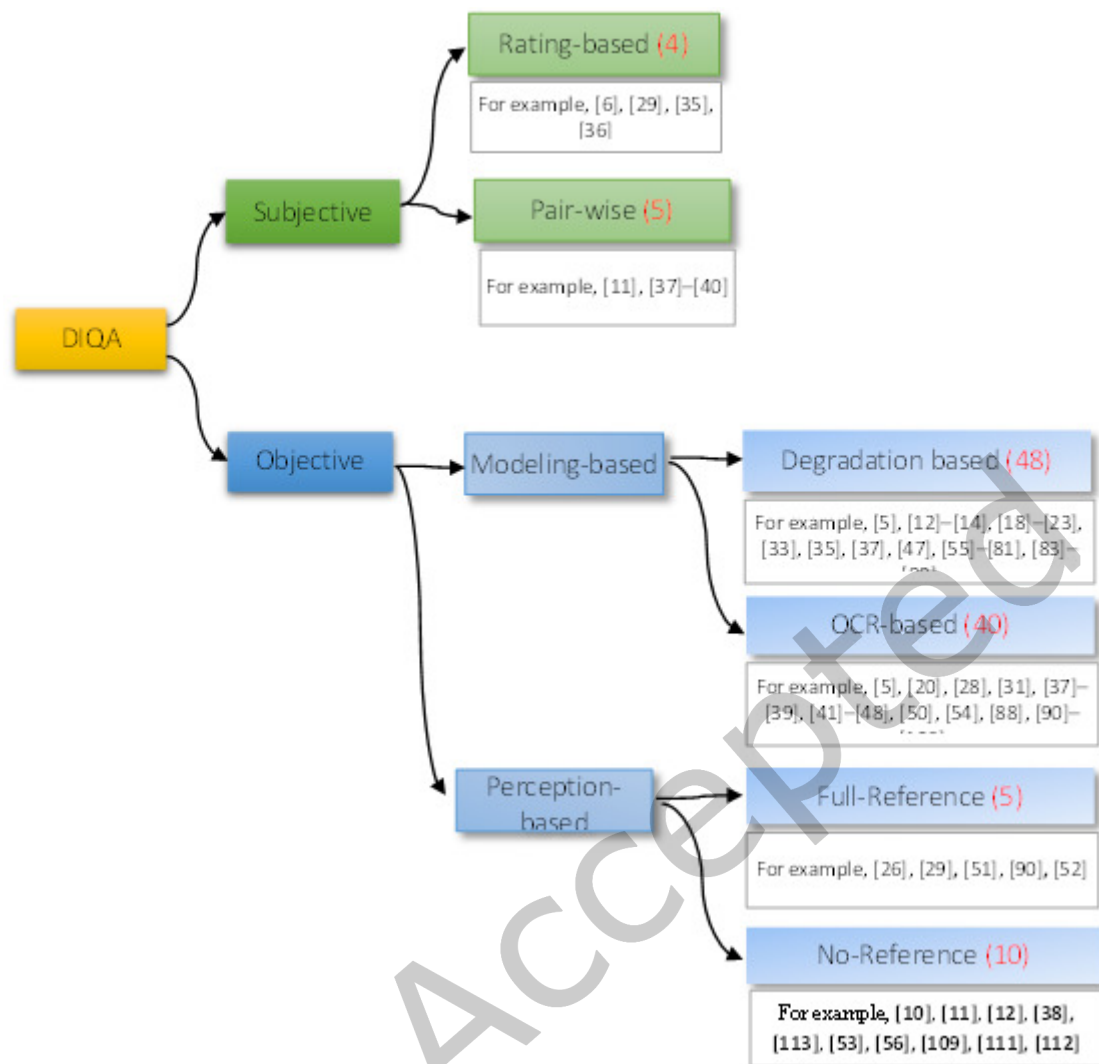


Figure 5: Hierarchical classification of DIQA methods in the literature.

A categorical representation of different methods in the literature and the number of papers in each category is also provided in Figure 5 to give the readers a quick reference to the state-of-the-art DIQA methods and the number of articles in the related literature. Following the hierarchical classification of the DIQA methods in Figure 5, a detailed description of each category, the methods, characteristics of the methods, and applications are further provided in the subsequent sections.

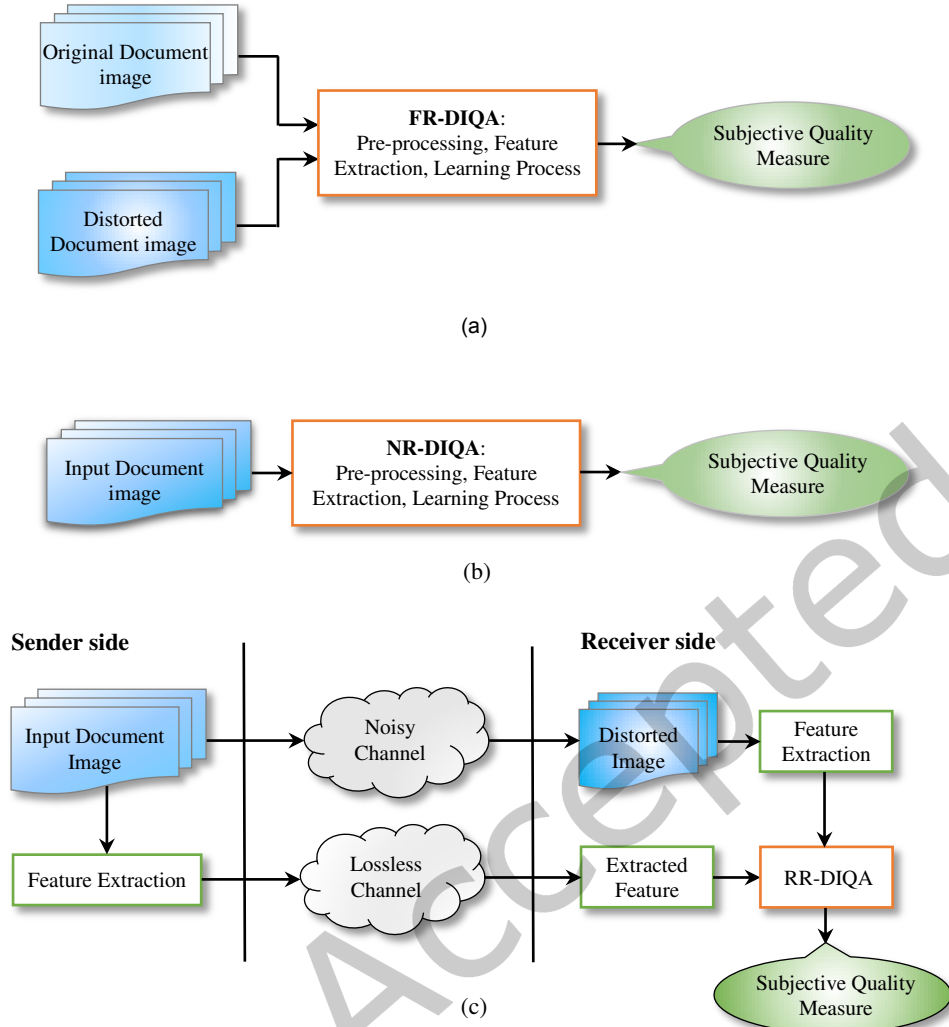


Figure 6: Three different general frameworks for (a) FR-DIQA, (b) NR-DIQA and (c) RR-DIQA.

3 SUBJECTIVE DIQA

Assessing the quality of an image by individuals' perception based on the human visual system is subjective. Different individuals may provide other perceptual quality measurements for an image, as human assessors have different views and opinions when looking at an image. Single-image ratings-based and pair-wise comparison-based approaches have been introduced in the literature to obtain subjective image quality assessment.

3.1 Ratings-Based Methods

In the ratings-based approach, subjects/assessors assess the quality of each image by assigning a categorical label to the given image. Four widely used nominal, ordinal, interval, and ratio measurement scales have been considered for subjective IQA [14]. A nominal scale describes an image's quality with categories that have no explicit order or ranking. For example, Baird [35] considered five different types of quality: Images with degradations caused by scanning, Highly degraded document images, Background degraded document images, Slightly degraded document images, and Good document images. An ordinal scale, however, orders the values but does not necessarily imply a quantitative association. For example, Bad, Poor, Fair, Good and Excellent could be used as a five-category ordinal scale to assess image quality. For an interval-based scale, the order is essential, and the difference between two scale values is also implied. An example of the interval scale is an exam score scale divided into five ranges (0-20, 21-40, 41-60, 61-80, 81-100). A ratio scale has a clear definition for each value on the scale. For example, a ratio scale can define the

quality of an image within the range of values between 0 and 100. Each value has its meaning for a certain quality. It is worth noting that depending on the criteria and instructions provided to the assessors and their cultural background, image qualities assigned to a given image may vary significantly. Therefore, the Mean Opinion Score (MOS) has commonly been used as the rating scale for estimating the quality of images in the literature. The MOS is the average/mean image quality for a given image computed based on quality values provided by a group of assessors. Considering this technique, the authors in [6], [29], [36] obtained Opinion Scores (OSs) for document images by asking each individual to evaluate the quality of a document image on various scales, for example, a scale of 0 to 100.

3.2 Pairwise-Based Methods

Pairwise comparison-based approaches require assessors to compare two images, generally a reference image and its distorted image or two distorted images, to assess quality. A few research works in the literature have used a pairwise approach to estimate the quality of document images. Hale and Barney-Smith [37] adopted an electronic version of the graphic rating scale used for psychophysical experiments to assess the quality automatically based on a computer-based method. Two images of degraded characters (with a slider below them) were shown to assessors, who were asked to choose the preferred image among the two character images. Kumar et al. [38] introduced a page-level subjective DIQA method to evaluate document image quality. The developed method is based on a crowd-sourced setting in which assessors were instructed to compare two degraded document images and select the preferred image [38]. Like the Kumar et al. approach, Obafemi-Ajayi and Agam [39] designed a DIQA model to assess a pair of images with different qualities. The assessors were then asked to choose the one with better (or slightly better) quality or if they seemed to be of identical quality. Shakolaei et al. [11], [40] used a pair comparison rating (PCR) to assess the visual quality of document images.

As can be seen from the literature, predicting document image quality using human perception has attracted very few researchers resulting in only a small number of publicly available document image datasets, which have considered human opinion scores as ground truth for quality assessment. This is partly because subjective DIQA by humans is time-consuming and expensive. It is also impractical in real-time applications. There are still many open research problems in subjective DIQA [14] that can be summarized. First, most subjective DIQA experiments have been designed for the character level rather than the document/page level. Second, in all subjective DIQA experiments, only one specific type of degradation has been assessed, so we can rarely study the interaction between degradation types. Third, the design of subjective DIQA experiments to assess the overall subjective quality of a document image when multiple degradations exist in the image is still an open research problem. Fourth, it is unclear how to instruct assessors to obtain reliable opinion scores, as the HVS responses to degradations appearing in the foreground and background of images differ. In addition, assessors may judge the quality of an image based on various factors, including aesthetics, noise and degradations, objects present in the image, and content. Finally, in the pair-wise comparison-based DIQA methods, the number of comparisons may become very large when the number of images is significant. If the number of images considered for an experiment is n , the total number of comparisons will become $\frac{n \times (n-1)}{2}$, and as a result, designing a low-cost subjective DIQA experiment is challenging.

To handle the vast volume of document images generated with typical applications and overcome some of the challenges mentioned above, designing and developing automatic objective DIQA methods is a reasonable solution. An overview of the aim of DIQA methods in the literature is presented in the following section.

4 OBJECTIVE DIQA

Objective DIQA aims to design and develop computational models that can automatically predict input document images' qualities and provide quantitative metrics close to subjective scores attained from individuals. Like IQA, the first objective DIQA measure used the full reference mean squared error (MSE) metric computed by averaging the squared intensity differences of distorted and reference

image pixels along with the related quantity of peak signal-to-noise ratio (PSNR) [24]–[26]. However, these metrics are not very well matched to perceived visual quality [14], [24]–[27], [29]. Subsequently, various sophisticated objective DIQA measures have been developed in the past three decades to cope with different downstream processes found in document image analysis applications [10], [14]. These methods are categorized into two main groups—modeling-based and human perception-based – considering the conceptual definition of objective document image quality in each method. For modeling-based approaches, the consumers of document images are typically a downstream process, for example, OCR. As a result, document image quality is defined based on, for example, the OCR accuracy and DIQA metrics are factors that can be used to reliably predict OCR accuracy [5], [13], [41]–[47]. Meanwhile, in the human perception-based approaches, objective document image quality measures are computed according to human perception, as the end-users/clients of document images may be individuals (humans) [5], [10]–[12], [26], [38], [39], [48]–[55]. A detailed discussion of each approach is provided in the following subsections.

4.1 Document modeling-based methods

There are two main categories of modeling-based DIQA metrics - document image degradation modeling and OCR prediction modeling. The former approach quantifies the quality of document images by modeling image degradations. The hypothesis is that document images with higher degradation levels would have poorer quality and vice versa. However, the latter methods align document image quality scores with OCR accuracies, assuming that OCR accuracies on high-quality document images would be better than on low-quality document images. As discussed earlier, the choice and development of DIQA metrics depend on whether the downstream processing must be carried out based on human perception or pure machine-generated results. The following subsections review document image degradation modeling and OCR prediction modeling proposed in the literature.

4.1.1 Document Image Degradation Modeling

Document image degradation is a change of one or more typical properties of an image at different stages of its life cycle, including document creation, document digitization, and document image processing, for various reasons, such as physical, human, and transformation impacts [14]. Image degradations usually reduce the visual quality or the “readability” of document images compared to their original [12], [35]. The performances of document image analysis applications, such as document image archival and retrieval, OCR, and image classification, are often negatively affected by document image degradation. Therefore, understanding document image degradations is critical for developing effective restoration and enhancement algorithms [13]. It is worth mentioning that image degradation methods, especially document image degradation modeling, could easily warrant a survey.

Nevertheless, we briefly review document image degradation modeling in this section for completeness. Table 1 summarizes the literature’s degradation models, features, datasets, and metrics. Readers can refer to [35] and [56] for a detailed document image degradation modeling overview.

A document image can conventionally be decomposed into text and non-text components, and a text portion may further be decomposed into lines, words, and characters/strokes using a top-down approach, considering the characteristics of document images. Degradations may affect the quality of a document image at all these levels. For example, blurring artifacts may negatively affect the quality of camera-captured document images at the page, line, and character levels. To better comprehend the effects of degradations on document images, document image degradations can be categorized into three classes, page, line, and stroke. Document images may be degraded at the page level due to background noises, such as salt & pepper noise, ruled lines, clutter, show-through, bleed-through, and margin noise.

Moreover, shape deformations, including image warping, curling, skew, and slant, may degrade the quality of document images at the page level. At the line level, text lines may touch each other, and there may be inconsistencies in gaps and distance within and between lines. Text line skew and curve shape lines may cause line-level image degradations. A document image may be damaged at the stroke level due to touching, broken characters, and additive noises (dots and irregular shapes).

As discussed, different types of degradation may affect the performance of document image analysis algorithms. Therefore, it is essential to design methods to understand the types and levels of degradations and how they would affect document image quality [13]. Several methods have been proposed in the literature [35], [56]–[59] to model and find document image degradation parameters. In the early 90s, Baird [57] proposed a parametric defect model to treat local defects of document images occurring due to the printing and imaging process. He considered different parameters of printing and imaging, including text size, resolution, rotation, multiplicative scaling factors, translation offsets, jitter, blur, sensitivity, and binarization threshold in the proposed defect model [57]. Later, Baird [60] further presented different methods to model calibration and comprehensively analyzed the methods and their applications in document image analysis.

Moreover, he also proved that document image quality could be automatically and reliably distinguished using synthetic degraded document images generated by a degradation model [61]. He further utilized the proposed models to create an extensive publicly accessible database of character images with the ground truth and their defect model parameters to run simulation studies for pattern recognition and enhance document image classifiers' performance [62], [63]. These methods are suitable for document images containing pure text and will not work well on color document images with complex backgrounds.

Kanungo et al. [64] proposed global and local distortion models to handle perspective distortion, non-linear illumination effects, and perturbation in the document images' optical scanning and digitization processes. They further improved the models by adopting morphological modeling and considering the perspective geometry of optical systems [65]. In addition, Zheng and Kanungo [66] proposed a document image restoration approach using a morphological degradation model. The model enhanced degraded document images of ten different degradations generated based on their unique parameter set. Kanungo and Zheng further introduced an optimization method based on the Kolmogorov-Smirnov test and the Nelder-Mead algorithm to estimate the parameters of a page-level degradation model for binary document images [67].

Other attempts to model document image degradations include [68]–[71]. Sural and Das [68] employed a two-state Markov chain-based approach to model isolated pixel reversal and blurring degradations in document images. Shadows and bleed-through degradations have also been modeled using optical or motion blurring, geometric transformations, spatial sampling, and a virtual diffusion process in scanned and camera-captured document images [69]–[71]. A restoration method based on the degradation models has further improved image quality [69]–[71]. Pan et al. [72] proposed an effective L0-regularization method based on intensity and gradient features to deblur text images and enhance image quality. Capel and Zisserman [73] introduced a maximum likelihood estimation method using a maximum posterior and total variation norm to model various degradations, including geometric warping, optical blur, spatial sampling, and noise for super-resolution text. However, as the impacts of several degradations on a document are different from a single degradation impact, these degradation models may not be applicable where a document suffers from all or several of these degradations simultaneously.

Moreover, a few researchers attempted to model degradations and defects caused by specific digitization devices, including photocopiers [23], scanners [74], and printers [75]. Smith [74] proposed a method to find the variance between the original and the digitized characters based on the point spread function diameter and the binary intensity threshold. He developed a scanner parameter estimation technique using bi-level star chart scans [76]. Machine learning has also been used to overcome technical difficulties in modeling the degradation of scanned documents. For instance, Peng et al. [77] proposed an unsupervised learning approach using the visual bag of words to model the degradations of scanned documents. This type of approach can bypass the task of

designing features for specific image distortion. The visual words have been fed into latent Dirichlet allocation (LDA) for the "quality" topics learning to model different types of degradation. A regression model and the document's OCR scores have been utilized to predict OCR results as an indication of degradation degrees [77]. It is worth mentioning that as the type and quality of digitization devices and, consequently, the quality of the digitization process is improving rapidly, these models are not suitable for current digitization technology,

In addition to degradation modeling, researchers introduced different methods to detect and recover documents from degradation. Drira et al. [33] presented a partial differential equation (PDE) based method to deal with the problem of degraded characters by preserving singularities and edge continuities. The technique adds constraints to the Weickert coherence-enhancing diffusion filter (anisotropic diffusion) to prevent rounding corners in the filtering and image enhancement step [33]. This method, however, cannot handle degradations caused by gaps and significant discontinuities in characters. Garg and Chaudhury [58] proposed an automated approach to optimize document image preprocessing algorithms. Shahkolaei et al. [12] proposed a model based on Generalized Gaussian Distribution (GGD) and Asymmetric GGD (AGGD) features along with a Support Vector Machine (SVM) classifier to detect four different degradations, including paper translucency, readers' annotations, stains, and worn holes in document images. Tan et al. [22] extended Peng's work in this area by proposing an efficient restoration method based on 3D shape modeling to remove shading information from scanned document images. The shading has been characterized using four components: a proximal and moving light source, Lambertian reflection, nonuniform albedo distribution, and document skew. Practical models have then been built based on a 3D geometric model and a 3D optical model to reconstruct the 3D shape of a book surface. Scanned document images have been restored using deshading and dewarping models [22]. Lu and Doooms [19] further proposed a generic method to identify and remove some specific physical distortions from document images. They constructed a conditional random field (CRF) framework to identify distortions based on distorted pixels' intensity and spatial relation. A peak searching method has then been employed to obtain model parameters of the energy functions in the conditional probability. The original and physical noise pixels were classified by maximizing the conditional probability in the CRF model [19]. However, the problem of document image degradation classification will become more complex when documents contain various components, such as graphic shapes, mathematical equations, and chemical formulas, rather than only textual data. In such a scenario, the performances of these models will decline as some of the actual components will be classified as distortions.

Filtering methods alone and along with quality assessment have also been proposed in the literature to perform preprocessing on historical and degraded document images to enhance their quality and readability [5], [47], [78]–[83]. Eye-tracking technology has further been used to aggregate the reading behavior of a group of individuals for providing objective quality feedback [55]. In most of these studies [5], [55], [78]–[80], human perception is used to determine the quality enhancement in document images.

Evaluation of degradation models has also been the objective of several studies in the literature. For example, Kanungo et al. [65], [84]–[86] presented a few methods using different validation procedures based on dataset sample size, power, and distance functions to evaluate other document image degradation models. They introduced a statistical method based on a two-sample nonparametric permutation test to statistically validate any degradation models [84], [86]. Li et al. [87] also proposed a validation method that compares OCR errors induced by the degradation model and errors encountered using real scanned documents. The link between model parameters and human perception concerning the observable degradation features has further been studied in the document image degradation modeling literature [23], [37], [83]. A summary of some essential features and methods used in document image degradation modeling literature is provided in Table 1 to give readers a quick idea of the document image degradation models and their characteristics.

From the literature on document image degradation modeling, it can be summarized that the generalization of degradation models, the number of parameters of degradation models, the model parameter estimation, and degradation model validation are four critical factors in designing effective document image degradation models [7], [30]. Degradation models should easily be generalized so that the model can duplicate any degraded image.

Also, the distribution of generated defective images must be close to the distribution of real-world documents. Moreover, models must be tuned to work using a small number of parameters to create random artifacts—various restoration and quality enhancement algorithms to enhance the quality [88]. However, applying a general-purpose document quality enhancement technique to different document images may further degrade document images.

Table 1: Features and models used in document image degradation modeling literature.

Reference	Model	Features
[12]	SVM-based learning model	Generalized Gaussian Distribution (GGD) and Asymmetric GGD (AGGD) on two layers of the degraded image
[58]	Supervised Filter Learning	Local features: normalized raw image patches Global features: Local descriptors are encoded by a set of linear filters
[61]	Hamming distance, Nearest neighbor	Symbol size, spatial sampling rate (resolution), skew, x and y scale factors, x and y offsets, jitter (pixel distribution in bounding box), speckle, blurring, thresholding
[68]	Two-state Markov chain model	Burst method for propagating pixel-level in the 8, 16, and 24 neighbor pixels
[77]	Unsupervised learning (LDA)	Raw pixel values from small patches of size 20×20
[84]	Morphological and Bel Lab image degradation models	The function of distance, Size, sampling rate, skew, vertical and horizontal translation offsets, multiplicative scaling factors
[86]	Pixel inversion and blurring models	Pixel-level features based on distance transform and random pixel-flipping process

4.1.2 OCR Prediction Modeling

OCR is the engine and core processing component of most document image analysis applications, such as document image retrieval and classification. Therefore, these applications' performances directly depend on the OCR results and performance. Accordingly, the accuracy of any OCR system relies on the quality of document images and the style of people's handwriting [14], [88], [89]. As a result, several DIQA methods in the literature have considered OCR accuracy, as a metric, to quantify the quality of document images [5], [20], [28], [38], [39], [44], [54], [88], [90]–[95]. It is worth noting that most of the OCR-based DIQA methods in the literature have been applied to machine-printed document images sourced from scanners, still cameras, and mobile cameras [28], [37], [39], [41]–[47], [54], [88], [90]–[93], [96]. Some of the DIQA measures are suitable only for binary printed document images [42], [45], [47]. The OCR-based metrics allow applications using an OCR as a part of their pipelines to automatically filter out document images with OCR accuracies lower than a predefined threshold and to perform manual data entry or processing for those low-quality document images. OCR accuracies may be impacted by document degradations, handwriting styles (in handwritten documents), or both [14].

A summary of OCR-based DIQA methods in the literature is presented in Table 2. As shown in Table 2, OCR-based DIQA methods in the literature can be described in terms of learning models and features. Notably, learning models are grouped into a) statistical topological analysis or heuristic-based methods, b) distance-based methods, and c) machine learning-based methods.

In the statistical topological analysis or heuristic-based approach, character-based features, such as the number of strokes or holes, aspect ratio, gravity position, and the number of black pixels, have commonly been used to characterize document images to assess their quality. Metrics obtained using those features and their

correlations with OCR results have mainly been considered the statistical means of document image quality in most of the works in the literature. High correlation values indicate that the performances of the proposed methods/metrics are commendable.

Several features, including font size, stroke thickness, small speckle factor (SSF) – a measurement of the number of black background speckles, White Speckle Factor (WSF) – a measure of the degree to which strokes are fattened, Touching Character Factor (TCF) – a measurement of the degree to which neighboring characters touch and Broken Character Factor (BCF) – a measure of the degree to which strokes are broken based on Connected Components (CCs) analysis, heuristics and parameter tuning have been computed to estimate the OCR accuracy and consequently to choose the best restoration method for document enhancement [42], [45], [47].

In addition, a comprehensive set of features has been used in various DIQA methods to quantify the quality of document images based on OCR accuracies. For example, Kumar and Ramakrishnan [46] used uniformity, sharpness, continuity, noise, stroke width variance, pulse width ratio, transient pixels density, entropy, and components variance for the DIQA. Features, such as the Gradient [41], [91], [94], Laplacian [44], [94], Wavelet [94], and the average character/word height-width ratio [41] have also been used in the literature to predict the OCR accuracy. These features have further been considered to find the relation between the OCR results and out-of-focus blur degradations of camera-captured document images [41], [94]. Chou and Yu [84] extracted three features, including the stroke-density distribution of a Chinese character, histogram, and crossing count, to further create a metric for measuring the quality of handwritten Chinese characters. They then used the Bayes decision rule for quality estimation. According to them, a character is defined as good quality if it is similar to the average of a large volume of that character [48].

Considering the distance-based approach for DIQA in the literature, Lu et al. [90] presented a method based on the distance-reciprocal distortion measure of two pixels for binary document images at the character level. This method is an FR-DIQA based on the observation that the distance between two pixels plays a significant role in their mutual interference perceived by humans [90]. Zhu and Milanfar [97] also proposed a no-reference sharpness metric (Q) based on the singular value decomposition (SVD) of the local image gradient matrix that performs well on the parametrization of an image restoration algorithm. Shemiakina et al. [96] recently presented an NR-DIQA method based on the geometric distortions and probability of correct text recognition to deal with projectively distorted images. Estimating document image quality using all pixels (the entire input image) would be time-consuming and may cause problems in real-time applications, such as computational complexity. The method, therefore, analyses geometric distortions instead of pixel-wise analysis of the entire image to estimate image quality [98]. In addition, a sound-emitting DIQA (SEDIQA) metric based on OCR has been developed for visually impaired people to hear the quality of document images and capture appropriate images in terms of OCR correctness [31]. Entropy and Gradient features have been extracted to build the SEDIQA metric [31].

Supervised and unsupervised machine learning DIQA methods based on OCR results have been presented in the literature [39], [77], [54], [92], [93], [28], [95], [43], [99]. Ye and Doermann [43] proposed an unsupervised feature learning framework followed by a Support Vector Regressor (SVR) to learn a mapping from image features to image quality scores using OCR accuracy of gray-scale machine-printed English and Greek document images. Raw image patches were extracted from unlabeled images, and a clustering method was used to learn a dictionary and obtain image representations for NR-DIQA. For the OCR quality prediction on a given input image, a set of raw-image patches were extracted and encoded using the learned dictionary based on soft-assignment encoding with max pooling. Finally, a Support Vector Regressor (SVR) model was used to learn a mapping from image features to an image quality score [43]. Xu et al. [100] further applied a similar approach to DIQA, where differences between high-order statistics of local features and codewords are defined as global quality-aware features.

Considering the supervised approach presented in the literature [39], [54], [92], [93], [28], [95], [99], [101], Obafemi-Ajayi and Agam [39] proposed a neural network-based quality assessment for binary document images at the character level using different mathematical morphology, noise removal, and characteristic spatial features to estimate the qualities of character images. Bhowmik et al. [95] proposed a technique based on the bag of allograph patterns and the similarity measure of text components by template matching along with an SVR model to predict the OCR results as a quality measure. Peng et al. [92] further extended the concept of the bag of allographs and sparse representation using a bag of visual word features in conjunction with a regression-based machine learning approach to estimate the quality of document images considering OCR performance as ground truth. Furthermore, Kang et al. [93] proposed a two-layer deep learning approach based on a convolutional neural network (CNN) for document image quality assessment. This method initially divides an input image into several non-overlapping patches and then uses a CNN to estimate quality scores for selected patches. The patch scores are then averaged to obtain the overall quality score for the input document image [93]. Later, Lu and Doooms [28] proposed a two-stage deep CNN (DCNN) based transfer learning procedure followed by a fine-tuning process to transfer knowledge of natural scene image characterization obtained from a pre-trained DCNN for no-reference DIQA. Peng and Wang [99] extended the use of the Siamese network by using ResNet as the feature extractor to develop a DIQA method. Unlike the conventional approaches, multiple patches from the same degraded document image are fed into the neural network simultaneously to learn the quality of each patch and the relations between patches. New losses based on the Pearson Linear Correlation Coefficient (PLCC) and Spearman Rank Order Correlation Coefficient (SROCC) measures have also been introduced to increase linearity and monotonicity [99].

Furthermore, Li et al. [54] presented a three-layer deep neural network composed of encoder, decoder, and prediction layers to detect text lines and then find their quality using a new loss function. The overall text-related image quality is computed using the weighted averaging method as the pooling strategy on all qualities obtained from detected text lines [54]. Li et al. [102] also developed an attention-based Recurrent Neural Network (RNN) model for camera-based DIQA. They integrated CNN and RNN to capture spatial features for several glimpse regions within image patches. Reinforcement learning is used to pay attention to the salient part to find the location of glimpse regions. The quality of a document image is obtained by computing the average value of quality scores of all the sifted patches [102].

It is worth noting that handcrafted features, including entropy, gradient, strokes, speckle, CCs, fonts, and noise, characterizing characters or text, still apply to the DIQA problems dealing with low-resource scripts, limited digitized data and low research development. However, considering the research advances and the availability of large datasets in Latin and Kanji-based scripts, in addition to the availability of massive and considerably cheap computing power in today's research, deep learning-based features are alternate or a better choice for DIQA. It is also important to note that supervised methods are effective mostly where labeled data are available. In low-resource scripts, the creation of datasets and labeling is a challenging task, and this causes slow advancement in this area of research.

There are also methods in the literature for the evaluation of the quality of document image datasets composed of mainly handwritten characters [103], [104], [50], [105]. Datasets are usually used for training, testing, and evaluating document image processing systems. It is, therefore, essential to know the quality of the datasets used for experiments. Shinghal and Suen [103] proposed a quality estimation metric based on frequency density and distance measurements to rank various handwritten Chinese character image models. Moreover, Hase et al. [104], [105] proposed a quantitative entropy (variation entropy) based method to estimate the quality of handwritten character binary images aiming to evaluate the quality of a handwritten Japanese character image dataset using OCR results. Park et al. [50] extended this method by defining an average entropy for gray-scale handwriting document images to evaluate gray-scale document image datasets. Bui et al. [106] used handcrafted features to characterize the blurriness in document images. They used SVMs to classify distorted document images' quality based on their blurriness level [106]. Recently, Li et al. [101] proposed a deep-learning method to predict document image quality based on OCR accuracies. CNNs have been designed to predict the quality of text lines in a document image. The overall quality of the document image has finally been assessed with the

ensemble of all text line qualities [101]. It is worth mentioning that most methods for handwritten document image quality assessment have been developed at the character level and may not work directly at the word and text levels.

Many methods in the literature of OCR-based DIQA work only on binary document images having entirely textual data [39]. They also focus on character-level quality assessment, far beyond current real-world scenarios. There is a lack of methods for handwritten document image quality estimation. In handwritten document image analysis, image quality and readability may differ. There are also only a few methods for the quality assessment of datasets. Methods for evaluating the quality of datasets, especially page-level handwritten document datasets, would be desirable for designing more accurate document image analysis systems. Moreover, one of the drawbacks of most of the existing DIQA methods is the use of hand-crafted features, which may work well only in specific contexts. Applying more sophisticated supervised and unsupervised (deep) feature learning and sparse coding-based methods may help address some of those issues [28], [77].

Table 2: OCR-based DIQA models and features used in the literature.

Reference	Learning Model	Features
[5]	Linear regression	Fragmentation (Number of pixels, Average width of CC, Average height of CC, Number of CC, Pixel density), Skew, Cut-of-document
[31]	Rule-based or decision tree	Entropy and gradient features
[39]	MLP feed-forward neural, MLP regression	6 Morphological-based features (dilation, erosion, opening, closing), 5 Noise-removal-based features (median and Gaussian filters), 10 Spatial characteristics features (foreground percent, gradient, CC, Length of segments)
[41]	Support Vector Regression (SVR)	The gradient of the edge, Average height-width ratio
[42]	Rule-based or decision tree	Font Size Factor, Small Speckle Factor, Stroke Thickness, White Speckle Factor, Touching Character Factor, Broken Character Factor
[43]	K-means clustering and SVR with linear kernel	Raw-image patches and pixel intensity as local descriptors
[44]	Metric-based approach	Laplacian based features
[45]	Rule-based or decision tree	White Speckle Factor, Broken Character Factor, Number of CC
[46]	Straight-line fitting using least squares and correlation coefficient	Uniformity, sharpness, continuity, noise, stroke width variance, pulse width ratio, transient pixels density, entropy and variance of components, Transient Region Density, Stability of CC values, Continuity, Noise measure based on Median Filtering, Pulse width ratio, Entropy
[47]	Linear classifier using the Pocket algorithm	Normalized Font Size, Small Speckle Factor, White Speckle Factor, Touching Character Factor, Broken Character Factor
[48]	Bayes decision rule	Stroke density distribution, Histogram, Crossing Count
[54]	Deep neural network and weighted pooling	Textline image patches

[83]	Correlation-based analysis	Font Size Factor (FSF), Stroke Thickness Factor (STF), Touching Character Factor (TCF), Small Speckle Factor (SSF), Broken Character Factor (BCF), White Speckle Factor (WSF)
[90]	Distance measure and nearest neighbor classifier	Pixel intensity
[91]	Metric based	Character gradient features
[92]	Semi-supervised sparse learning	Bag-of-visual-words (BoV) based features
[93]	Deep neural network	Image patches
[28]	Transfer learning-based deep neural network	Image patches of size 227×227
[94]	Metric-based	Fusion of focus measure operators such as Gradient-(GRA), Laplacian-(LAP), and Wavelet-based (WAV) operators
[96]	Singular value decomposition no-reference sharpness metric	Local image gradient matrix
[100]	Unsupervised feature learning using K-means clustering and SVR with linear kernel	Raw-image patches and pixel intensity as local descriptors and differences between high-order statistics of local features and codewords as global features
[99]	Siamese deep neural network	Image patches of size 128×128
[102]	Transfer learning-based deep neural network	Image patches of size 256×256
[105]	Information criterion learning	Variation Entropy (variation entropy for a unit boundary length and variation entropy for a unit area)
[101]	CNN based	Raw pixels extracted from text lines
[106]	SVM based	Handcrafted features characterizing the blur issue

4.2 Human Perception-based Methods

Unlike OCR-based, human perception-based DIQA can assess both textual and non-textual documents. Earlier research on human perception-based image quality assessment (IQA) has focused on natural scene images [107], [108]. This is due to the massive number of daily natural scene images and videos generated. Like natural scene images, a considerable volume of document images is generated every day. However, natural scene images (NSIs) and document images, including screen content images (SCIs), are different as cameras generally capture NSIs from real-world sceneries. At the same time, computer graphics may initially design document images, and they are captured from papers and computers' (or any other devices') screens [92], [109]. In addition, structure, statistics, visual appearance, visual attention, and degradation impacts of document and natural scene images are different. Document images may contain simple text as a foreground and fancy decoration and graphical content with complex foreground and background patterns. They may be stylish and colorful with an aesthetically pleasing outlook. Moreover, foreground and background in document images may degrade differently [12], [40]. As a result, general IQA methods may not perform well in the case of document images [11], [12], [26], [28], [29], [53], [88], [109]–[112].

Though several OCR-based DIQA methods are presented in the literature [14], OCR-based methods may fail to provide reliable quality assessment results for these types of documents. Despite being a good quality image from a human perspective, OCR has failed to perform well in several cases. Moreover, the OCR-based approach requires running the full preprocessing, segmentation, and recognition workflow on the entire document image. This process imposes unnecessary processes, which are time-consuming and error-prone, causing problems in mobile and embedded applications [98].

Therefore, in the past two decades, some human perception-based DIQA methods have been developed to automatically estimate the quality of document images before the employment of any further document image processing [7], [37], [49], [51], [111]. A common approach in human perception-based DIQA methods is to use MOSs as the ground truth to train automatic quality assessment models and then evaluate them to see how objective image qualities computed by a machine are close to the image quality scores of individuals. Based on the availability of the original/reference images, DIQA methods presented in the literature can be categorized into two main groups, including (i) full reference (FR) and (ii) no reference (NR) or blind DIQA.

FR-DIQA is the case where reference images are available and used to compute image qualities. NR- or blind DIAQ estimates quality scores for document images without using original or reference images. A list of FR- and NR-DIQA methods, along with their features and learning methods, are shown in Table 3. The subsequent subsections provide a detailed discussion of FR- and NR-DIQA methods.

4.2.1 Full-Reference DIQA (FR-DIQA)

FR-DIQA methods compute image qualities by comparing original/reference document images with their distorted versions [26], [51], [52], [90]. FR-IQA methods have been designed based on the assumption that structural information changes in distorted images compared to their original ones can provide an excellent approximation to perceived image distortion [25]. FR DIQA methods have also benefited from the structural similarity principle, as documents generally contain highly structural contents [109], [110]. FR-DIQA methods in the literature employed various features characterizing the textural, visual, and structural appearance of document images to estimate the quality of document images. Considering the FR-DIQA methodologies, almost all the FR-DIQA methods in the literature belong to the metric-based approach.

Metric-based FR-DIQA methods used different features, such as the gradient, texture, and Hough derivations, extracted from document images to create various quality maps, including gradient and quality maps, to estimate the qualities of the input document images [26], [51], [52]. For instance, Alaei et al. [26] proposed a method to extract a set of foreground patches from a reference image. For each foreground patch, its gradient magnitude similarity map was computed using the original gradient magnitude map and its corresponding distorted one. Gradient magnitude similarity deviation of the patch is calculated, and an average pooling is finally performed to obtain the final image quality of the distorted document image [26]. The same authors further proposed an FR-DIQA method based on texture features using local binary patterns (LBP) to extract LBP features at local and global levels of each document image to compute a similarity measure called the LBP similarity index (LBPSI) [51]. The local and global LBP features are combined using a weighting strategy to compute the LBPSI and achieve the final FR-DIQA results [51].

Moreover, Alaei [52] presented another FR-DIQA based on the first and second-order Hough derivations to create a similarity map from reference and distorted images. An average pooling is then employed to obtain a quality score for the distorted document image [52]. The main issue with the gradient magnitude, LBP, and Hough features is that they primarily represent local changes at a point in document images and cannot capture the global information, which is also required for DIQA.

In addition, an FR method for assessing screen content image quality using sharpness and luminance features and a weighted similarity between the reference and distorted screen images is proposed [29].

To our knowledge, there is no FR-DIQA method based on deep learning in the literature. Machine learning-based FR-DIQA methods generally learn a mapping model between features extracted from a set of images (or

image patches) and their corresponding subjective quality scores in the training phase [14]. This can be a possible future avenue for further DIQA research.

Table 3: The literature uses human perception-based DIQA methods, features, and learning models.

Reference	Type	Learning Model	Features
[26]	FR-DIQA	Metric-based approach: Gradient Magnitude Similarity Deviation	Gradient magnitude
[29]	FR-DIQA	Metric-based approach: Weighted similarity index	Sharpness and luminance features
[51]	FR-DIQA	Metric-based approach: LBP Similarity Index	Local binary pattern (LBP) features
[52]	FR-DIQA	Metric-based approach: Hast Similarity Index	Hast derivative features
[10]	NR-DIQA	Modification of quality-aware clustering method: k-mean clustering and distance-based approach	Difference of Gaussian (DoG) features, Gradient Magnitude and Phase Congruency features
[11]	NR-DIQA	Generalized Gaussian distribution and asymmetric Gaussian distribution	Statistics of the mean subtracted contrast normalized (MSCN) coefficients and Gabor filter analysis
[12]	NR-DIQA	Logistic regression-based model	Generalized Gaussian Distribution (GGD) and Asymmetric GGD (AGGD) on two layers of the degraded image
[38]	NR-DIQA	Rule-based or decision tree classifier and regression-based model	Sharpness features (difference of grey-scale values of a median-filtered image)
[53]	NR-DIQA	SVR based model	Two sets of features of free energy measure and structural degradation information
[111]	NR-DIQA	SVR based model	Four types of features: picture complexity, screen content statistics, global brightness quality, and sharpness
[112]	NR-DIQA	SVR based model	A set of learned filters to utilize both local and global features

4.2.2 No-Reference DIQA (NR-DIQA)

Since, in most real-world applications, original or reference images are not available, assessing the quality of a document image must be performed based on the image itself without considering any reference images. As a result, NR-DIQA is the most desirable solution for estimating the qualities of document images. Notably, most methods in the literature of objective DIQA are based on the NR approach. Distortion-specific and general-purpose NR-DIQA algorithms can be developed to assess

document image quality. It is worth mentioning that almost all NR-DIQA methods in the literature have used learning-based models to estimate document image qualities.

The learning-based approach for DIQA has better performance than other DIQA techniques. Perceptual-based blind document image quality metrics, such as Modification of Quality Aware Clustering (MQAC), filter learning, and Multi-distortion Document Quality Measure (MDQM), have been proposed in the literature [10]–[12], [38], [112]. Kumar et al. [38] proposed an NR-DIQA method to estimate the blurriness of document images that occurred due to the camera motion (or hand-shake), defocus, or inherent properties of the imaging system. The sharpness/blurriness of edges has been estimated based on the changes in greyscale (luminance) values observed at edges [38]. Moreover, Alaei et al. [10] proposed a foreground/background segmentation technique followed by a patch selection strategy integrated into a bag-of-visual-words-based patch image quality representation. A full reference image quality assessment method followed by the k-mean clustering technique was applied to obtain the bag-of-visual-words patch image quality. The document image was first divided into a set of patches. The extracted patches have then been filtered out, and their qualities have been estimated using a clustering assignment. An average pooling has finally been employed to assess the quality of the input document image [10]. Later, Shahkolaei et al. [11] used the statistics of the mean subtracted contrast normalized (MSCN) coefficients obtained from segmented layers of each document image to compute another objective NR-DIQA metric called Visual DIQA Metric (VDIQAM). Log-Gabor filters were applied to segment a document image into four foreground and background layers to extract attractive HVS features [11]. They further proposed a regression-based MDQM method to assess the quality of physically degraded document images based on MOSs [12]. Three sets of spatial and frequency image features extracted from two layers of text and non-text regions of document images were used to regulate the proposed MOS-based regression model for DIQA [12]. In addition, an NR-DIQA based on filter learning and regression modeling has been proposed by Ye et al. [112]. As a compact set of filters, local and global features have been extracted using the back-projection method and supervised learning [112].

Recently, Courtney [32] used the concept of discrete entropy (DE) and proposed an NR quality metric to deal with scanned document images. The DE metric may not capture well contrast, smoothness, and sharpness from document images, resulting in poor quality assessment, Gu et al. [53] further proposed a blind quality metric based on Support Vector Regression (SVR) and 13 different structural degradation and free energy features for screen content images (SCIs). The model has been trained using 100,000 SCIs and their associated machine-generated quality scores and provided good results compared to subjective quality scores. Moreover, they extracted four types of features, including picture complexity, screen content statistics, global brightness quality, and sharpness from screen content [111]. They then trained a regression model using many samples labeled with a high-performance FR screen content IQA method [111]. Tao et al. [109] further differentiated between natural scene images (NSIs) and screen content images (SCIs), as cameras generally capture NSIs from the real-world scene. At the same time, SCIs are typically a mixture of pictures and text. They proposed a non-uniform scale adaptive sampling model (NUSM) to preprocess the input images enabling traditional IQA methods to perform well on SCIs [109]. To deal with identity cards, as another form of physical documents, Yanez and Juan used the blind/referenceless image spatial quality evaluator (BRISQUE) method to train a new model and assess the quality of Chilean national ID cards [36]. Schulz et al. [113] developed an NR method for assessing the quality of ID Card images by combining face and text IQA. The IQA using a quality-aware face recognition method was further used to help Presentation Attack Detection (PAD) in ID cards PAD systems [114]. The application of IQA for PAD is in its early stage and requires more investigation.

Although a few NR-DIQA algorithms have been presented in the literature, NR-DIQA intrinsic difficulty and the issues of NR-DIQA are still challenging and remain largely unexplored to date.

5 EVALUATION METRICS, BENCHMARKS, AND DATASETS

This section summarizes evaluation metrics, benchmarks, and datasets used in DIQA methods. Evaluation metrics are used in DIQA methods to measure how close subjective and objective estimated values are. The literature's three commonly used evaluation metrics are the SROCC, the PLCC, and the Kendall Rank Correlation Coefficient (KRCC). The PLCC and SROCC metrics

measure the prediction linearity and monotonicity of two sets of values. The KRCC measures the degree of similarity between estimated quality scores and MOS. An excellent objective quality assessment method should typically achieve high values (close to 100%) in SROCC, PLCC, and KRCC metrics. Moreover, the Root Mean Square Error (RMSE) has also been used as the evaluation metric to assess DIQA methods. This metric is the prediction error standard deviation, indicating how concentrated the data is around the best fit line. This value should generally be close to 0 for good objective quality assessment methods. To evaluate OCR-based DIQA methods in the literature, accuracy (*Acc*) and Levenshtein distance (*Lev*) have been used as the most common evaluation metrics. The following are the formulas to compute the metrics mentioned above.

$$PLCC = \frac{n(\sum_1^n x_i y_i) - (\sum_1^n x_i)(\sum_1^n y_i)}{\sqrt{[n \sum_1^n x_i^2 - (\sum_1^n x_i)^2][n \sum_1^n y_i^2 - (\sum_1^n y_i)^2]}} \quad (1)$$

In Equation 1, n is the number of samples in the dataset, and x_i and y_i are tagged and predicted qualities of sample i , respectively.

$$SROCC = 1 - \frac{6 \sum_1^n d_i^2}{n(n^2 - 1)} \quad (2)$$

In Equation 2, n is the number of samples in the dataset, and d is the difference in the ranks of the tagged (x_i) and predicted (y_i) qualities of sample i .

$$KRCC = \frac{(C - D)}{(C + D)} \quad (3)$$

In Equation 3, C and D are the total numbers of concordant and discordant pairs of ranks of tagged and predicted qualities, respectively.

$$RMSE = \sqrt{\frac{\sum_1^n (x_i - y_i)^2}{n}} \quad (4)$$

In Equation 4, n is the number of samples in the dataset, and x_i and y_i are tagged and predicted qualities of sample i , respectively.

$$Acc = \frac{k}{n} \quad (5)$$

The *Acc* in a multiclass classification problem is computed by dividing the number of correct classifications (k) by the total number of samples (n) in the dataset.

$$Lev(a, b) = \begin{cases} \begin{matrix} |a| & \text{if } |b|=0 \\ |b| & \text{if } |a|=0 \end{matrix} & \text{if } a(0) = b(0) \\ 1 + \min \begin{cases} Lev(tail(a), b) \\ Lev(a, tail(b)) \\ Lev(tail(a), tail(b)) \end{cases} & \text{otherwise} \end{cases} \quad (6)$$

The Levenshtein distance, also called *edit distance*, is a metric for measuring the difference between two strings, and it is the minimum number of single-character edits (insertions, deletions, or substitutions) required to change one string into the other. The Levenshtein distance, $Lev(a, b)$, of two strings a and b of lengths $|a|$ and $|b|$, respectively, is calculated using equation 6, where the *tail* of the string x is a string of all but the first character of x , and $x[n]$ is the n^{th} character of the string x , counting from 0.

Datasets are fundamental in DIQA research as they provide a common ground for research in this area. As there are two main classes of DIQA methods (OCR- and perception-based methods), two corresponding categories of datasets have been created and used in the literature [38], [42], [66], [96], [115].

Blando et al. [45] created one of the first OCR-based datasets in the literature. It comprises two subsets of data containing 460 book pages and 200 magazine pages chosen from 100 magazines published in the U.S. in 1992. Document pages have been digitized in 300 dpi resolution and then binarized. Six OCR tools have been used to annotate subsets, and character accuracy has been computed for each image [45]. Cannon et al. [47] created a dataset of 139 type-written document images of 300 dpi resolution [47]. OmniPage Pro v8.0 has been used to perform OCR, and the character error rate of the corpus is 20.27%. They created a sub-corpus of 41 highly degraded document images with OCR character error rates between 20% and 50%. They also created a small corpus of documents spanning a range of gradually decreasing quality by repeatedly photocopying a page from a book (a total of 9 versions). Each successive copy has been degraded with background speckles, widened stroke widths, touching characters, and other typical attributes of lower-quality document images [47].

Zheng and Kanungo [66] further created a dataset of 100 one-column printed pages of the English Bible. The 100 images have been degraded and categorized into ten groups based on their unique parameter set. FineReader 4.0 has been used for OCRing, and OCR accuracy error rates have been computed at the character and word levels as ground truth [66]. Souza et al. [42] used another dataset comprising 736 printed English document images with various font sizes, types, and styles. Each document image contains only one text line without graphics, tables, drawings, or underlined text. Almost all images suffer from some degradation, such as broken characters, touching characters, salt-and-pepper noise, or a combination of two or more problems [42]. Moreover, Later, Kumar et al. [96] created a public dataset called SOC composed of 175 camera-captured document images containing various levels of focal blur introduced manually during the capturing times. The images are generated using 25 documents from two publicly available datasets, the University of Washington Dataset and TobaccoDatabase. Six to eight images have been taken from a fixed distance for each document to capture the whole page. Three recent OCR methods have been applied to obtain the quality of images in terms of character-level OCR results [96].

Zi and Doermann [70], [71] presented a document image degradation methodology by incorporating various noises at the page and pixel levels. Ground truth and degraded images have been generated automatically from the electronic text, which can be used in training or evaluating document analysis systems. Peng et al. [41] created a dataset of 235 high-quality two-tone Arabic text document images for DIQA. Four degraded images (three out-of-focus blur images and one auto-focus feature) using a digital camera have been generated. The WER (Word Error) for each document has been calculated using BBN's OCR engine and considered the document image's quality score. The dataset is not publicly available [41]. Antonacopoulos et al. [116] created a dataset of 740 printed English and Greek text zone images from a collection of grayscale newspaper images. OCR accuracy at the character level obtained from FineReader 9 for each document image has been used to indicate document quality [116]. Similarly, Ye and Doermann [43] constructed a dataset of 521 text zone images with more than 30 characters per document. The OCR accuracy for each document image has been computed for OCR quality prediction [43].

Nayef et al. [117] introduced the Smartdoc-QA dataset, composed of 4,260 document images of three different types, including modern documents, receipts, and old administrative letters. The document images have been captured using smartphone cameras with varying lighting conditions, blur degrees, and perspective angles. The dataset consists of different ground truths, including the documents' text transcriptions, the documents' OCR results, and the values of the various capture parameters used for each image. This dataset is suitable for evaluating OCR-based quality assessment methods' performance and quality enhancement methods [117]. To get an idea of the Smartdoc-QA dataset, a sample document from this dataset and its corresponding low-quality image is shown in Figure 7.

Lorillard
 LORELLARD, INC. Research Center, 420 Eighth Street, P.O. Box 21588, Greensboro, North Carolina 27402-1588

December 3, 1986

Dr. R. M. David
 Microbiological Associates, Inc.
 5271 River Rd.
 Bethesda, MD 20816

Dear Ray:

Enclosed are the signed protocols for the Phage Induction Studies: 7-7023.401 (A165), 7-7024.401 (A176) and 7-7025.401 (A178). However, beginning with the next set of protocols, we would like the following statement added to Section 7.2: "Tissues from all surviving animals will be used for assay."

If you have any questions, please feel free to call.

Sincerely,
Thomas A. Vollmuth
 Thomas A. Vollmuth, Ph.D.
 Toxicologist, Life Sciences
 TAV/jwl

Lorillard
 LORELLARD, INC. Research Center, 420 Eighth Street, P.O. Box 21588, Greensboro, North Carolina 27402-1588

December 3, 1986

Dr. R. M. David
 Microbiological Associates, Inc.
 5271 River Rd.
 Bethesda, MD 20816

Dear Ray:

Enclosed are the signed protocols for the Phage Induction Studies: 7-7023.401 (A165), 7-7024.401 (A176) and 7-7025.401 (A178). However, beginning with the next set of protocols, we would like the following statement added to Section 7.2: "Tissues from all surviving animals will be used for assay."

If you have any questions, please feel free to call.

Sincerely,
Thomas A. Vollmuth
 Thomas A. Vollmuth, Ph.D.
 Toxicologist, Life Sciences
 TAV/jwl

(a)

(b)

Figure 7: An original image from the Smartdoc-QA dataset (a) and one of its low-quality versions.

Recently, Yang et al. [29] created the Screen Image Quality Assessment Database (SIQAD) dataset, which is composed of two separate parts, including 20 original screen content images (SCIs) and their associated 980 distorted images. SCIs are high-quality “webpage” and “screensnap” images downloaded from the “Google Images” website. Each SCI has been distorted by using seven distortion types (Gaussian noise (GN), Gaussian blur (GB), motion blur (MB), contrast change (CC), JPEG and JPEG2000 (JP2K) compressions, and Layer Segmentation based Coding (LSC)) at seven degradation levels. The images have been scored for quality by human experts. The second part contains 1000 high-quality “webpage” and “screensnap” images downloaded from the “Google Images” website. From 1000 original SCIs, a set of 100,000 distorted images has been created using the above distortions at different levels. A state-of-the-art method has generated a quality score for each distorted image. Therefore machine-generated scores have been considered ground truth or labels of those 100,000 distorted images [29].

Considering human perception as ground truth, Kumar and Ramakrishnan [46] created a database of 132 annotated document images written in Indian scripts comprised of various degradations. Each document image was annotated by six individuals on a scale between 1 and 5, and the average of human scores was considered the quality score of each document image [46]. In addition, Kumar et al. [38] created a perception-based dataset called DocSharp corpus from a magazine and technical article document images taken by smartphone users at various viewing angles and distances from the document. A total of 135 images have been collected and labeled based on pairwise comparison to find the sharpness (most in focus) image of a pair of images by 21-25 judges/workers at Amazon’s Mechanical Turk [38].

Alaei et al. [6], [26] created a human perception-based dataset called the ITESOFT to study the different aspects of document image quality and readability. The ITESOFT dataset contains 29 reference document images collected from real-world data using mobile cameras and scanners. Six different JPEG and JPEG2000 (JPEG2K) compression levels have been applied to the original images to generate 348 (29×2×6) distorted images. In total, 377 (348+29) document images with Mean Opinion Scores (MOSs) for the quality and readability of document images have been collected in the ITESOFT dataset. MOSs have been collected by asking a group of people to evaluate the quality and readability of a document image between 0 and 100 based on a set of criteria [6], [26].

Shahkolaei et al. [11] created a Multi-distortion Historical Document Image Database (MHDID) comprising 335 historical document images of 1024 * 1280 and their associated MOSs. The MHDID document images have been categorized into four groups based on their distortion types, including paper translucency (88 images), stains (113 images), readers' annotations (61 images), and worn holes (73 images). The visual qualities of degraded ancient document images have been assessed by 15 assessors using the pair comparison rating and the MOS values. The dataset can be used for document image quality assessment and degradation classification [40]. The same group of researchers applied the same procedures to create a visual document image quality assessment (VDIQA) dataset [11], [40]. The differences between the MHDID and VDIQA are the number of images in the datasets and the number of assessors. The VDIQA contains only 177 document images of 400 * 2000 with MOS values obtained from 10 assessors [11], [40].

Yanez and Juan [36] created a private dataset of 204 ID card images captured from three different Chilean ID cards (digital, printed, and screen), with 68 images in each category. Images are of size 320 × 240 pixels and have different qualities, ranging from completely out of focus or tampered to well-focused or real-looking forgeries. This team [113] further created the ID-CARD-CLv2 dataset and images were captured using smartphone and tablet models for a remote verification system. The dataset consists of 12,960 images of Chilean national identity cards with various resolutions along with 12,960 images of face selfies corresponding to the same subjects from the ID cards.

Table 4 summarizes the datasets and their characteristics used in the literature for DIQA. Table 4 and the literature show that machine-printed character datasets were initially created, and DIQA methods primarily focused on the quality of characters. Later, DIQA progressed to page-level document images. It is also observed that most datasets are limited to scanned document images. There are a limited number of camera-based document image datasets for OCR quality assessment in the literature. The first dataset [41] comprises only Arabic documents and is not publicly available. The second one comprises English documents with 6–8 different degradations per document freely available [96]. However, the degradations and distortions associated with camera-captured images are very different from scanned images.

Although a significant amount of research has been carried out to develop computational models and create standard datasets for natural scene image quality assessment, quality assessment of document images, especially the human perception-based approach, needs more attention and further research. From the literature, it can be noted that there are a minimal number of human perception-based DIQA datasets publicly available. In addition, there is no principled and straightforward way to conduct subjective DIQA experiments. Therefore, it is necessary to construct document image quality datasets with subjective labels provided by humans and make them publicly available to the research community.

Table 4: Datasets used in the literature for DIQA.

Reference {year}	Description	Type/ Granularity	Language	Ground Truth	Use
[11] {2018}	177 pages http://www.synchromedia.ca/system/files/VDIQA.zip (Not accessible)	Color/Page	Arabic	Human opinion scores	Perception-based DIQA and degradation classification
ITESOFT [26] {2015}	29 original, 248 degraded (Private dataset)	Color/ Page	French	Human opinion scores	Perception-based DIQA and degradation classification
DocSharp	135 iPhone and Android pages	Color/	English	Human opinion scores by	Perception-based DIQA (only)

[38] {2012}	(Not available)	Page		Amazon's Mechanical Turk	sharpness/ blurriness)	
[41] {2011}	235 printed and handwritten words and 940 corresponding degraded word images (Not available)	Binary/Word Character	Arabic	OCR accuracy	OCR-based assessment	quality
[42] {2003}	736 printed characters (370 for training, 183 for validation, 183 for testing) (Not available)	Binary/ Character	English	3 OCR accuracies	OCR-based assessment and filter selection	quality
[43] {2012}	740 newspaper text zone images (Not available)	Gray/ Character	English and Greek	OCR accuracy	OCR-based assessment	quality
[45] {1995}	660 1 document images (21 for training, 639 for testing) (Not available)	Binary/ Character	English	6 OCR accuracies	OCR-based assessment	quality
[46] {2011}	132 document pages (Not available)	Color/Page	Multi- scripts (Tamil and Kannada)	Human scores opinion	Perception-based DIQA	
[47] {1999}	139 printed character images (Not available)	Binary/ Character	English	OCR accuracy	OCR-based assessment	quality
[62] {1991}	2500 printed character images (Not available)	Binary/ Character	English	OCR accuracy	Degradation modeling and classification	
[66] {2001}	100 document images (Not available)	Binary/Page	English	OCR accuracy	Degradation modeling and classification, OCR- based DIQA	
[70] {2005}	18,144 printed word images (Not available)	Binary/word	English, Chinese, Farsi, Urdu, Arabic	OCR accuracy	Degradation modeling and OCR- based quality assessment	
[40] {2018}	335 historical document images http://www.synchromedia.ca/system/files/VDIQA.zip (Not accessible)	Color/Page	Arabic	Human scores opinion	Perception-based DIQA and degradation classification	
SOC [96] {2013}	175 document images (Not available)	Gray/Charact er	English	3 OCR accuracies	OCR-based assessment	quality

[101]	52,094 synthesized text lines	Gray/Line	English, Chinese	Machine-generated quality scores	OCR-based quality assessment	quality
{2019}	(Not available)					
Smartdoc-QA [117]	4,260 document images http://navidomass.univ-lr.fr/SmartDoc-QA	Color/Page	English, French	2 accuracies	OCR	OCR-based quality assessment
{2015}						
[36]	204 images of ID cards	Color/Page	Spanish	Human opinion scores	Perception-based DIQA	
{2021}	(Not available)					
ID-CARD-CLv2 [113]	12,960 images of ID cards (Not available)	Color/Page	Spanish	Human opinion scores	Perception-based DIQA	
{2022}						
SIQAD [29]	1,000 screen content images	Color/Page	English	Human opinion scores	Perception-based Screen Content IQA	
{2015}	https://sites.google.com/site/guke198701/publications					

6 COMPARISON OF THE STATE-OF-THE-ART METHODS

Results from different state-of-the-art methods concerning document image degradation modeling, OCR-based and FR and NR perception-based DIQA are summarized in Tables 6, 7, and 8, respectively. It is important to note that the results reported in terms of different evaluation metrics, including PLCC, SRCC, and OCR accuracy scores, should be considered with caution, as they may not be directly comparable and have not been accompanied by statistical significance test results.

The results shown in Table 5 show that degradation models in the literature have been evaluated and benchmarked with different datasets. Moreover, there is no common or standard evaluation protocol and metrics. Thus, the results shown in Table 5 are not comparable. Furthermore, as each method is designed for a specific degradation, general degradation models cannot handle different degradations. The literature shows that most models are proposed for scanned binary images and are not suitable for mobile and camera-based color document images.

Table 5: Results in the literature of document image degradation modeling.

Reference	Dataset	Type of result	Results
[12]	MHDID [40]: 335 document images	Degradation Modeling	85.1% degradation classification accuracy
[58]	500 English paragraphs	Degradation Modeling	PLCC = 98.4% SROCC = 97.9%
	470 Hindi paragraphs	Degradation Modeling	PLCC = 88.5% SROCC = 78.8%
	540 Bangla paragraphs	Degradation Modeling	PLCC = 73.4% SROCC = 70.1%
[61]	Synthetic binary document images 25 images for evaluation	Degradation Modeling	Reliability based on statistical tests and visual inspection
[77]	SOC dataset [96]: 102 for training and 73 samples for testing	Degradation Modeling	PLCC = 91.3%
[84]	Real and synthetic binary document images of printed characters	Degradation Modeling	Reliability-based on statistical tests
[86]	Real and synthetic binary document images of printed characters	Degradation Modeling	Reliability-based on statistical tests
[5]	i) 574 pages out of 1415 English forms scanned in gray-scale @ 212 DPI and binarized. ii) 876 document images	OCR	OCR Acc. = 90.0%
[39]	3190 binary-level character images Training: 11 English characters and 20-40 samples for each Testing: 220 images (20 samples for 11 characters)	OCR	PLCC = 85.5%

From the results of OCR-based DIQA methods shown in Table 6, we can observe a growth in technology and standardization of evaluation protocols and metrics. However, a direct comparison of the DIQA methods reported in the literature until 2010 is impossible as most researchers used their own built datasets and evaluation protocols. After 2010 with the SOC dataset in 2013 [96] and the Smartdoc-QA dataset in 2015 [117], a direct comparison of some OCR-based DIQA methods became possible thanks to standard datasets and the use of PLCC and SROCC metrics. From the results reported based on the SOC dataset [96] in Table 6 (continue), it is clear that the method proposed by Lu and Doots [28] provides the best performance in terms of PLCC (96.30%) and SROCC (93.10%) metrics. The same comparison could be performed considering the Smartdoc-QA dataset [117]. However, the evaluation protocols used for experimental analysis are not the same.

The perception-based DIQA results in Table 7 reveal that PLCC, SROCC, RMSE, and PSNR are the most frequent evaluation metrics used in FR-/NR-DIQA methods in the literature. VDIQ and MHDID [40] are the

two recently created datasets used in most newly proposed methods. ITESOFT dataset has also been used for FR- and NR-DIQA. Alaei et al. [51] obtained the best PLCC and SROCC results when considering the ITESOFT dataset for experimental analysis [51]. It is also noted that perception-based DIQA methods have used available natural scenes datasets, such as LIVE [118], CSIQ [119], and TID2008 [120], for experiments.

Table 6: OCR-based DIQA results reported in the literature based on various datasets.

Reference	Dataset			Results	
	Data type	Training Data	Testing Data		
[47]	Corpus of gray-level, 300 dpi printed English document images	139 images	N/A	Reducing OCR error from 20.27% to 12.60%	
[41]	Machine-printed scanned and binarised Arabic text documents	120 original + 480 degraded images	115 original + 460 degraded images	PSNR = 7.18 for Test 4	
[42]	Printed binary English document images	370/183 images	183 images	OCR Acc. = 73.24-93.09% (%)	
[43]	Text zone images from a collection of greyscale newspaper images with printed English and Greek.	219 images	521 images	PLCC = 77.54%	RMSE = 14.17
[45]	460 pages of ISRI's printed scientific and technical English documents scanned in binary at 300 dpi.	21 pages	439 pages	PLCC = 86.80%	
[45]	200 pages of magazines scanned at 300 dpi.	N/A	N/A	PLCC = 86.50%	
[83]	32 frequent English word images with different degradation parameters (567 instances for each). 18,144 word images	N/A	N/A	Correlation between degradations and features (OCR results)	
[90]	60000 small text binary images	N/A	N/A	PLCC = 96.40%	
[93]	521 grayscale newspaper images with various resolution	60000 training + 20000 validation patches	20000 test patches	PLCC = 73.00%	SROCC = 73.00%
[100]	A total of 521 text zone images from a collection of grayscale newspaper images with printed English and Greek	N/A	N/A	PLCC = 80.00%	SROCC = 76.60%
				RMSE = 14.2	

Table 6 (Continue): OCR-based DIQA results reported in the literature based on the SOC and Smartdoc-QA datasets.

Reference	Dataset			Results	
	Data type	Training Data	Testing Data	PLCC (%)	SROCC (%)
[44]	SOC dataset [96]: 175	80%	20%	88.70	82.70

[54]	color document images	60%	40%	96.00	92.90
[91]		80%	20%	90.60	85.70
[92]		80%	20%	93.50	92.80
[93]		90,000 training, 30,000 validation	30,000	95.00	89.80
[28]		80%	20%	96.50	93.10
[94]		N/A	N/A	93.80	96.40
[97]		N/A	N/A	82.70	93.70
[100]		N/A	N/A	96.00	90.90
[99]		140	35	95.70	93.60
[102]		80%	20%	95.60	91.60
[101]		Synthesized data [101]	Entire SOC	90.89	87.23
[112]		80%	20%	92.60	85.40
[31]	Smartdoc-QA dataset	NA	Smartdoc-QA	84.63	85.82
[54]	[117]: 4260 color document images	60%	40%	68.40	67.30
[99]		80% for training and 10% for validation	10%	92.70	78.80
[102]		140	35	81.40	86.50
[101]		Synthesized data [101]	Smartdoc-QA	74.33	75.72

Table 7: Perception FR- and NR-DIQA results reported in the literature considering various datasets and metrics.

Reference	Dataset	Results				
	Data type	Type of result	PLCC (%)	SROCC (%)	RMSE	PSNR
[26]	LIVE dataset [118]: 808 natural scene images	FR-DIQA	94.6 0	95.1 0	8.82	
[51]			94.0 0	93.8 0	9.34	
[52]			94.2 0	94.7 0	9.15	
[26]	CSIQ dataset [119]: 896 natural scene images		94.5 0	94.9 0	0.86	
[29]	SIQAS: 980 screen content images		85.8	84.1	7.34	11.59

			4	6		
[26]	ITESOFT dataset: 377 color document images		91.7	91.6	0.09	
			0	0	6	
[51]			93.1	91.9	0.08	
			0	0	8	
[52]			92.2	91.4	0.09	
			0	0	4	
[10]		NR-DIQA	89.1	87.5	0.10	
			0	0	9	
[10]	LIVE dataset [118]: 808 natural scene images	NR-DIQA	90.2	90.1	11.6	
			0	0	9	
[53]			87.0	89.0		
			0	0		
[112]			92.1	92.0		0.867
[10]	CSIQ dataset [119]: 896 natural scene images		89.2	84.9	0.12	
			0	0	8	
[53]			86.0	84.0		
			0	0		
[10]	TID2008 dataset [120]: 1725 natural scene images		86.7	84.9	0.79	
			0	0		
[112]			86.8	83.0		0.832
[11]	VDIQA [40]: 177 degraded historical Arabic document images (80% for training, 20% for testing)		84.8	81.7		64.92
			1	6		
[12]			87.6	84.4		67.85
			8	7		
[11]	MHDID [40]: 335 degraded historical Arabic document images (80% for training, 20% for testing)		82.9	79.6		61.15
			0	0		
[12]			86.6	83.9		65.66
			7	5		
[111]	SIQAS: 980 screen content images		79.0	76.2	8.75	11.59
			6	5	56.65	

7 DISCUSSION, CHALLENGES, AND REMARKS

We have observed from the DIQA and document image degradation modeling literature that most currently available DIQA methods use OCR as the quality metric [44], [93]. The number of DIQA methods based on human perception is relatively small compared to general-purpose IQA methods in the literature. Moreover, there are still many challenges in the domain of DIQA, especially in the human perception-based DIQA approach. This suggests many open research problems that need to be addressed by researchers. Few DIQA

methods based on human perception have also been developed to deal with color and grayscale document images [14], [38]. In addition, most methods have been designed for a specific type, for example, two-tone document images or a particular level of granularity in document images, such as page and individual character levels. As many document images are captured in color or grayscale, existing technologies for quality assessment of two-tone images may not perform well in some applications. Furthermore, many current DIQA methods have specifically been developed to estimate only one type of distortion, such as blur, and cannot be generalized. However, a typical degraded image may contain various degradations at different levels. It is also worth noting that it is unclear how to obtain page-level DIQA for document images containing multiple degradations.

The literature has also shown that foreground information or distortion attracts greater human attention than background information or distortion [121]. A proposal based on a fusion approach using different layers of data with different weights can also be investigated to see the impact of different layers of information.

Another open issue is that document image ‘quality’ and ‘readability’ are typically closely linked. A good (poor) quality document image is a readable (unreadable) document. This assumption may be accurate for machine-printed document images containing mostly text [5], [6]. Still, other document images, such as handwritten and CAPTCHA documents, are difficult for machines to deal with but can easily be read by humans. Therefore, these two terms may need to be redefined and investigated by researchers to better understand the relation between them.

Concerning features used for model-based DIQA in the literature, different derivative features, including Gradient, Laplacian, Hough, the difference of Gaussian, and sharpness, have been extracted from document images and provided promising DIQA results [10], [26], [38], [44], [52]. The application of other features in the DIQA literature can further be investigated. In addition, the deep learning approach is an emerging technology that has not been widely used and explored in DIQA literature. To our knowledge, there is no FR-DIQA method based on deep CNN learning in the literature. However, a few deep learning-based methods for NR-DIQA concerning OCR and perception scores exist. Therefore, DIQA techniques based on deep learning features, which do not require employing handcrafted feature extraction methods, can be a new direction for more research and exploration.

It is also noted that the methods based on machine learning have demonstrated encouraging results for accurately estimating the quality of document images. The learning-based approach can easily be generalized and is less sensitive to the training data. However, this method generally needs enough labeled data to train the models. To the best of our knowledge, there are no sufficiently large standard datasets that researchers can use as benchmarking datasets to compare the performances of DIQA methods. The availability of such datasets will help to promote and grow the research in the domain of DIQA. The OCR-based labeled data can be generated with minimal human effort. However, human perception-based labeling needs many individuals involved in this process which is tedious, time-consuming, and expensive. A standard dataset based on MOS could, therefore, be used to significantly extend DIQA research [10], [38], [88]. This can help apply perception-based DIQA approaches in applications, such as digital libraries, to understand and measure to what extent their digitization aligns with human expectations and needs.

The same observation applies to the need for a very large OCR-based labeled dataset for DIQA. In the creation of new DIQA datasets, including uniform distribution of images with different qualities (balance dataset), representativeness of the domain (context), independence from OCR products, completeness in terms of the presence of noises and degradations, and correctness of ground truth, are some essential factors that should be considered. These factors are also vital in developing end-to-end DIQA frameworks where the quality of the ground truth dramatically affects the performance of the DIQA models. In addition, research work in the literature studied the comparison and relationships between OCR and PSNR, considering the impact of JPEG compression. It has been reported that the relation between OCR accuracy and PSNR is non-linear [122]. However, to the best of our knowledge, there is no research and dataset with both OCR and human opinion scores, as two different types of ground truth, to deeply explore and investigate the relationships between OCR

and perceptual quality in document images. This may further address the relationships between machine and human document readability.

Moreover, it is important to note that the choice between OCR and perception-based DIQA methods depends entirely on the applications. In some applications, for example, text search and banking, the OCR is a crucial part of the system. Therefore, an OCR-based DIQA would be a better choice to estimate image quality. However, in some other applications, such as historical document archives, human perceptual-based DIQA methods would be preferred to do the job as OCR may fail in such applications. Hybrid methods, by applying/combining both OCR and perception-based approaches, may provide better outcomes. The hybrid approach has not been explored in the literature of DIQA and would be a possible direction for future research. Moreover, considering the applications of face image quality assessment (FIQA), it is worth exploring and investigating the hybrid approach by combining a DIQA and FIQA to assess the quality of, for example, ID card, passport, and travel document images as another future research direction.

DIQA metrics are usually used in real-time document image analysis applications and/or communication systems. Therefore, the time complexity efficiency of DIQA methods is another important factor in designing document quality assessment metrics. Document images are generally of large sizes, for example, 2160×8840 pixels. In addition, a massive volume of documents is generated and transferred daily. These two parameters, image size, and volume, require any proposed DIQA metrics to be computationally fast and efficient. It has been found that, for example, using smaller patch sizes takes a longer time to extract features resulting in slower systems [112]. On the other hand, using a patch selection strategy can increase the execution speed of the system [10]. Though a few DIQA methods in the literature have considered and discussed this point in their design [10]–[12], [98], [112], future investigation is required to fully explore what features and models can perform the best considering the conditions mentioned above.

REFERENCES

- [1] D. Doermann, E. Rivlin, and A. Rosenfeld. 1998. "The function of documents. *Image Vis. Comput.*, 16, 11, pp. 799–814, doi: [https://doi.org/10.1016/S0262-8856\(98\)00068-7](https://doi.org/10.1016/S0262-8856(98)00068-7).
- [2] "International Standards ISO/TC 42." 2022. https://www.imaging.org/site/IST/Standards/TC_42_Standards/IST/Standards/TC42.aspx?hkey=208a8a91-14da-4b6a-a92d-7637238308e4 (accessed Sep. 23, 2022).
- [3] "ISO 20462-1:2005 Photography — Psychophysical experimental methods for estimating image quality — Part 1: Overview of psychophysical elements." 2022. <https://www.iso.org/standard/38330.html> (accessed Sep. 23, 2022).
- [4] A. Mittal, A. K. Moorthy, A. C. Bovik, and L. K. Cormack. 2011. "Automatic prediction of saliency on JPEG distorted images. In *Proceedings of the International Workshop on Quality of Multimedia Experience*, pp. 195–200.
- [5] V. Govindaraju and S. N. Srihari. 1995. "Image quality and readability. In *Proceedings of the International Conference on Image Processing (ICIP)*, pp. 324–327. doi: 10.1109/ICIP.1995.538547.
- [6] A. Alaei, R. Raveaux, D. Conte, and B. Stantic. 2018. "'Quality' vs. 'Readability' in document images: Statistical analysis of human perception. In *Proceedings of the International Workshop on Document Analysis Systems (DAS)*, pp. 363–368.
- [7] L. Hu, Z. Hu, P. Bauer, T. Harris, and J. Allebach. 2020. "Document image quality assessment with relaying reference to determine Minimum readable resolution for compression. *Electron. Imaging*, pp. 323-1-323-8, 2020.
- [8] A. Radford and K. Narasimhan. 2018. Improving Language Understanding by Generative Pre-Training. in *Computer Science*.
- [9] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. 2018. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *ArXiv Prepr. ArXiv181004805*.
- [10] A. Alaei, D. Conte, M. Martineau, and R. Raveaux. 2018. Blind document image quality prediction based on a modification of quality-aware clustering method integrating a patch selection strategy. *Expert Syst. Appl.*, 108, pp. 183–192, doi: doi.org/10.1016/j.eswa.2018.05.007.
- [11] A. Shahkolaei, H. Z. Nafchi, S. Al-Maadeed, and M. Cheriet. 2018. Subjective and objective quality assessment of degraded document images. *J. Cult. Herit.*, 30, pp. 199–209, doi: doi.org/10.1016/j.culher.2017.10.001.
- [12] A. Shahkolaei, A. Beghdadi, and M. Cheriet. 2019. Blind quality assessment metric and degradation classification for degraded document images. *Signal Process. Image Commun.*, 76, pp. 11–21.
- [13] R. T. Hartley and K. Crumpton. 1999. Quality of OCR for degraded text images. In *Proceedings of ACM conference on Digital libraries*, pp. 228–229. doi: doi.org/10.1145/313238.313387.
- [14] P. Ye and D. Doermann. 2013. Document image quality assessment: A brief survey. In *Proceedings of International Conference on Document Analysis and Recognition (ICDAR13)*, pp. 723–727.
- [15] E. N. Dalal et al.. 2003. INCITS W1. 1 standards for perceptual evaluation of text and line quality. In *Proceedings of Image Quality and System Performance, International Society for Optics and Photonics*, pp. 34–38.
- [16] S. Athar and Z. Wang. 2019. A comprehensive performance evaluation of image quality assessment algorithms. *IEEE Access*, 7, pp. 140030–140070, doi: 10.1109/ACCESS.2019.2943319.
- [17] T. Schlett, C. Rathgeb, O. Henniger, J. Galbally, J. Fierrez, and C. Busch. 2021. Face image quality assessment: A literature survey. *ACM Comput. Surv.*, doi: 10.1145/3507901.
- [18] R. D. Lins. 2009. A taxonomy for noise in images of paper documents-the physical noises. In *Proceedings of International Conference*

Image Analysis and Recognition, Springer, pp. 844–854.

- [19] T. Lu and A. Doms. 2018. Towards physical distortion identification and removal in document images. In *Proceedings of the European Workshop on Visual Information Processing (EUVIP)*, pp. 1–6. doi: 10.1109/EUVIP.2018.8611786.
- [20] R. D. Lins, B. Miro, and G. P. e Silva. 2009. An OCR assessment of the quality of document images acquired with portable digital cameras. Presented at the *Camera based Document Analysis and Recognition*, Barcelona, Spain, pp. 1–5.
- [21] Lins, Rafael Dueire, G. P. e Silva, and A. R. G. e Silva. 2007. Assessing and improving the quality of document images acquired with portable digital cameras. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR07)*, pp. 569–573.
- [22] C. L. Tan, L. Zhang, Z. Zhang, and T. Xia. 2006. Restoring warped document images through 3D shape modeling. *IEEE Trans. Pattern Anal. Mach. Intell.*, 28, 2, pp. 195–208, doi: 10.1109/TPAMI.2006.40.
- [23] E. B. Smith and X. Qiu. 2003. Statistical image differences, degradation features, and character distance metrics. *Int. J. Doc. Anal. Recognit. IJDAR*, 6, 3.
- [24] Z. Wang, A. C. Bovik, and E. P. Simoncelli. 2005. Structural approaches to image quality assessment. *Handb. Image Video Process.*, 7, p. 18.
- [25] Zhou Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli. 2004. Image quality assessment: from error visibility to structural similarity. *IEEE Trans. Image Process.*, 13, 4, pp. 600–612, doi: 10.1109/TIP.2003.819861.
- [26] A. Alaei, D. Conte, and R. Raveaux. 2015. Document image quality assessment based on improved gradient magnitude similarity deviation. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR15)*, pp. 176–180.
- [27] Z. Wang and A. C. Bovik. 2009. Mean squared error: Love it or leave it? A new look at signal fidelity measures. *IEEE Signal Process. Mag.*, 26, 1, pp. 98–117, doi: 10.1109/MSP.2008.930649.
- [28] T. Lu and A. Doms. 2019. A deep transfer learning approach to document image quality assessment. In *Proceedings of the 2019 International Conference on Document Analysis and Recognition (ICDAR19)*, pp. 1372–1377.
- [29] H. Yang, Y. Fang, and W. Lin. 2015. Perceptual Quality Assessment of Screen Content Images. *IEEE Trans. Image Process.*, 24, 11, pp. 4408–4421, doi: 10.1109/TIP.2015.2465145.
- [30] P. Singh, E. Vats, and A. Hast. 2018. Learning surrogate models of document image quality metrics for automated document image processing. In *Proceedings of the International Workshop on Document Analysis Systems (DAS)*, pp. 67–72. doi: 10.1109/DAS.2018.14.
- [31] J. Courtney. 2021. SEDIQA: Sound emitting document image quality assessment in a reading aid for the visually impaired. *J. Imaging*, 7, 9, p. 168, doi: 10.3390/jimaging7090168.
- [32] J. Courtney. 2020. CleanPage: Fast and Clean Document and Whiteboard Capture. *J. Imaging*, 6, 10, doi: 10.3390/jimaging6100102.
- [33] F. Drira, F. LeBourgeois, and H. Emptoz. 2012. A new PDE-based approach for singularity-preserving regularization: application to degraded characters restoration. *Int. J. Doc. Anal. Recognit.*, 15, 3, pp. 183–212, doi: 10.1007/s10032-011-0165-5.
- [34] N. Nayef, J. Chazalon, P. Gomez-Krämer, and J.-M. Ogier. 2014. Efficient Example-Based Super-Resolution of Single Text Images Based on Selective Patch Processing. In *Proceedings of the International Workshop on Document Analysis Systems (DAS)*, pp. 227–231. doi: 10.1109/DAS.2014.25.
- [35] H. S. Baird. 2007. The state of the art of document image degradation modelling. In *Proceedings of the Digital Document Processing. Advances in Pattern Recognition*, Springer, pp. 261–279.
- [36] C. Yáñez and J. Tapia. 2021. Image quality assessment on identity documents. In *Proceedings of the International Conference of the Biometrics Special Interest Group (BIOSIG)*, pp. 1–5. doi: 10.1109/BIOSIG52210.2021.9548294.
- [37] C. Hale and E. Barney-Smith. 2007. Human image preference and document degradation models. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR07)*, pp. 257–261.
- [38] J. Kumar, F. Chen, and D. Doermann. 2012. Sharpness estimation for document and scene images. In *Proceedings of International Conference on Pattern Recognition (ICPR)*, pp. 3292–3295.
- [39] T. Obafemi-Ajayi and G. Agam. 2011. Character-based automated human perception quality assessment in document images. *IEEE Trans. Syst. Man Cybern.-Part Syst. Hum.*, 42, 3.
- [40] A. Shahkolaei, A. Beghdadi, S. Al-Maadeed, and M. Cheriet. 2018. MHDID: A multi-distortion historical document image database. In *Proceedings of the International Workshop on Arabic and Derived Script Analysis and Recognition (ASAR)*, pp. 156–160.
- [41] X. Peng, H. Cao, K. Subramanian, R. Prasad, and P. Natarajan. 2011. Automated image quality assessment for camera-captured OCR. In *Proceedings of the International Conference on Image Processing (ICIP09)*, pp. 2621–2624.
- [42] A. Souza, M. Cheriet, S. Naoi, and C. Y. Suen. 2003. Automatic filter selection using image quality assessment. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR03)*, pp. 508–512.
- [43] P. Ye and D. Doermann. 2012. Learning features for predicting OCR accuracy. In *Proceedings of the International Conference on Pattern Recognition (ICPR)*, pp. 3204–3207.
- [44] N. Nayef and J.-M. Ogier. 2015. Metric-based no-reference quality assessment of heterogeneous document images. In *Proceedings of the Document Recognition and Retrieval XXII, International Society for Optics and Photonics*, p. 94020L.
- [45] L. R. Blando, J. Kanai, and T. A. Nartker. 1995. Prediction of OCR accuracy using simple image features. In *Proceedings of the International Conference On Document Analysis and Recognition (ICDAR95)*, pp. 319–322.
- [46] D. Kumar and A. Ramakrishnan. 2011. QUAD: Quality assessment of documents. In *Proceedings of the International Workshop on Camera based Document Analysis and Recognition*, pp. 79–84.
- [47] M. Cannon, J. Hochberg, and P. Kelly. 1999. Quality assessment and restoration of typewritten document images. *Int. J. Doc. Anal. Recognit. IJDAR*, 2, 2–3.
- [48] S.-L. Chou and S.-S. Yu. 1993. Sorting qualities of handwritten Chinese characters for setting up a research database. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR93)*, pp. 474–477.
- [49] V. Kulesh, K. Schaffer, I. Sethi, and M. Schwartz. 2001. Handwriting quality evaluation. In *Proceedings of the International Conference on Advances in Pattern Recognition*, Springer, pp. 157–165.
- [50] J.-S. Park, H.-J. Kang, and S.-W. Lee. 2000. Automatic quality measurement of gray-scale handwriting based on extended average entropy. In *Proceedings of the International Conference on Pattern Recognition (ICPR)*, pp. 426–429.
- [51] A. Alaei, D. Conte, M. Blumenstein, and R. Raveaux. 2016. Document image quality assessment based on texture similarity index. In *Proceedings of the International Workshop on Document Analysis Systems (DAS)*, pp. 132–137.
- [52] A. Alaei. 2019. A new document image quality assessment method based on Hast derivations. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR19)*, pp. 1244–1249. doi: 10.1109/ICDAR.2019.00201.
- [53] K. Gu, G. Zhai, W. Lin, X. Yang, and W. Zhang. 2016. Learning a blind quality evaluation engine of screen content images.

Neurocomputing, 196, pp. 140–149.

- [54] H. Li, J. Qiu, and F. Zhu. 2018. TextNet for text-related image quality assessment. In *Proceedings of the International Conference on Artificial Neural Networks*, Springer, pp. 275–285.
- [55] R. Biedert, A. Dengel, M. Elshamy, and G. Buscher. 2012. Towards robust gaze-based objective quality measures for text. In *Proceedings of the Symposium on Eye Tracking Research and Applications*, pp. 201–204.
- [56] M. Cheriet and R. F. Moghaddam. 2008. DIAR: Advances in degradation modeling and processing. In *Proceedings of the International Conference Image Analysis and Recognition*, Springer, pp. 1–10.
- [57] H. S. Baird. 1992. Document image defect models. In *Proceedings of the Structured Document Image Analysis*, Springer, pp. 546–556.
- [58] R. Garg and S. Chaudhury. 2016. Automatic selection of parameters for document image enhancement using image quality assessment. In *Proceedings of the International Workshop on Document Analysis Systems (DAS)*, pp. 422–427.
- [59] T. K. Ho and H. S. Baird. 1997. Large-scale simulation studies in image pattern recognition. *IEEE Trans. Pattern Anal. Mach. Intell.*, 19, 10, pp. 1067–1079.
- [60] H. S. Baird. 1993. Calibration of document image defect models. In *Proceedings of the Annual Symposium on Document Analysis and Information Retrieval*, pp. 1–16.
- [61] H. S. Baird. 1999. Document image quality: Making fine discriminations. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR99)*, pp. 459–462.
- [62] H. S. Baird. 1991. A 100-font classifier. In *Proceedings of the International Conference On Document Analysis and Recognition (ICDAR91)*, St.-Malo, France, pp. 1–13.
- [63] H. S. Baird. 1993. Document image defect models and their uses. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR93)*, pp. 62–67.
- [64] T. Kanungo, R. M. Haralick, and I. Phillips. 1993. Global and local document degradation models. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR93)*, pp. 730–734.
- [65] T. Kanungo. 1996. Document degradation models and a methodology for degradation parameter validation. Ph.D. Thesis, University of Washington, Seattle.
- [66] Q. Zheng and T. Kanungo. 2001. Morphological degradation models and their use in document image restoration. In *Proceedings of the International Conference on Image Processing (ICIP)*, pp. 193–196.
- [67] T. Kanungo and Q. Zheng. 2004. Estimating degradation model parameters using neighborhood pattern distributions: An optimization approach. *IEEE Trans. Pattern Anal. Mach. Intell.*, 26, 4, pp. 520–524.
- [68] S. Sural and P. Das. 1999. A two-state Markov chain model of degraded document images. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR99)*, pp. 463–466.
- [69] R. F. Moghaddam and M. Cheriet. 2009. Low quality document image modeling and enhancement. *Int. J. Doc. Anal. Recognit. IJDAR*, 11, 4.
- [70] G. Zi. 2005. GroundTruth generation and document image degradation. Master thesis, University of Maryland, College Park. [Online]. Available: <https://drum.lib.umd.edu/handle/1903/2524>
- [71] G. Zi and D. Doermann. 2005. Groundtruth generation and document image degradation. Maryland University, College Park, Department of Computer Science.
- [72] J. Pan, Z. Hu, Z. Su, and M.-H. Yang. 2017. L0-Regularized intensity and gradient prior for deblurring text images and beyond. *IEEE Trans. Pattern Anal. Mach. Intell.*, 39, 2, pp. 342–355, doi: 10.1109/TPAMI.2016.2551244.
- [73] D. Capel and A. Zisserman. 2000. Super-resolution enhancement of text image sequences. In *Proceedings of the International Conference on Pattern Recognition (ICPR)*, pp. 600–605.
- [74] E. H. B. Smith. 1998. Characterization of image degradation caused by scanning. *Pattern Recognit. Lett.*, 19, 13.
- [75] M. Norris and E. H. Barney Smith. 2004. Printer modeling for document imaging. In *Proceedings of the International Conference on Imaging Science, Systems, and Technology (CISST04)*, Las Vegas, NV, pp. 14–20.
- [76] E. H. B. Smith. 2001. Scanner parameter estimation using bilevel scans of star charts. In *Proceedings of the International Conference On Document Analysis and Recognition (ICDAR01)*, pp. 1164–1168.
- [77] X. Peng, H. Cao, and P. Natarajan. 2015. Document image OCR accuracy prediction via latent Dirichlet allocation. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR15)*, pp. 771–775.
- [78] G. Agam, G. Bal, G. Frieder, and O. Frieder. 2007. Degraded document image enhancement. In *Proceedings of the Document Recognition and Retrieval XIV, International Society for Optics and Photonics*, p. 65000C.
- [79] Z. Shi and V. Govindaraju. 2004. Historical document image enhancement using background light intensity normalization. In *Proceedings of the International Conference on Pattern Recognition (ICPR)*, pp. 473–476.
- [80] E. Kavallieratou and E. Stamatatos. 2006. Improving the quality of degraded document images. In *Proceedings of the International Conference on Document Image Analysis for Libraries (DIAL)*, pp. 10–pp.
- [81] Nikolaos Ntogas and Dimitrios Ventzas. 2009. Binarization of pre-filtered historical manuscripts images. *Int. J. Intell. Comput. Cybern.*, 2, 1, pp. 148–174, doi: 10.1108/17563780910939282.
- [82] W. A. Mustafa, S. Sam, M. A. Jamlos, and W. Khairunizam. 2021. Effect of different filtering techniques on medical and document image. In *Proceedings of the National Technical Seminar on Unmanned System Technology 2019*, Z. Md Zain, H. Ahmad, D. Pebrianti, M. Mustafa, N. R. H. Abdullah, R. Samad, and M. Mat Noh, Eds., Singapore: Springer Singapore, pp. 727–736.
- [83] D. K. Reed and E. H. B. Smith. 2008. Correlating degradation models and image quality metrics. In *Proceedings of the Document Recognition and Retrieval XV, International Society for Optics and Photonics*, p. 681508.
- [84] T. Kanungo, H. S. Baird, and R. M. Haralick. 1995. Validation and estimation of document degradation models. Presented at the symposium for document analysis and information retrieval, Las Vegas, NV, pp. 217–228.
- [85] T. Kanungo, R. M. Haralick, and H. S. Baird. 1995. Power functions and their use in selecting distance functions for document degradation model validation. In *Proceedings of the International Conference On Document Analysis and Recognition (ICDAR95)*, pp. 734–739.
- [86] T. Kanungo, R. M. Haralick, H. S. Baird, W. Stuezele, and D. Madigan. 2000. A statistical, nonparametric methodology for document degradation model validation. *IEEE Trans. Pattern Anal. Mach. Intell.*, 22, 11.
- [87] Y. Li, D. Lopresti, G. Nagy, and A. Tomkins. 1996. Validation of image defect models for optical character recognition. *IEEE Trans. Pattern Anal. Mach. Intell.*, 18, 2, pp. 99–107.
- [88] R. Walha, F. Drira, F. Lebourgeois, C. Garcia, and A. M. Alimi. 2018. Handling noise in textual image resolution enhancement using online and offline learned dictionaries. *Int. J. Doc. Anal. Recognit.*, 21, 1–2.

- [89] P. Sharma and S. Sharma. 2016. An analysis of vision based techniques for quality assessment and enhancement of camera-captured document images. In *Proceedings of the International Conference-Cloud System and Big Data Engineering*, pp. 425–428.
- [90] H. Lu, A. C. Kot, and Y. Q. Shi. 2004. Distance-reciprocal distortion measure for binary document images. *IEEE Signal Process. Lett.*, 11, 2.
- [91] H. Li, F. Zhu, and J. Qiu. 2018. CG-DIQA: No-reference document image quality assessment based on character gradient. In *Proceedings of the International Conference on Pattern Recognition (ICPR)*, pp. 3622–3626.
- [92] X. Peng, H. Cao, and P. Natarajan. 2016. Document image quality assessment using discriminative sparse representation. In *Proceedings of the International Workshop on Document Analysis Systems (DAS)*, pp. 227–232.
- [93] L. Kang, P. Ye, Y. Li, and D. Doermann. 2014. A deep learning approach to document image quality assessment. In *Proceedings of the 2014 IEEE International Conference on Image Processing (ICIP)*, pp. 2570–2574.
- [94] M. Rusiñol, J. Chazalon, and J.-M. Ogier. 2014. Combining focus measure operators to predict ocr accuracy in mobile-captured document images. In *Proceedings of the International Workshop on Document Analysis Systems (DAS)*, pp. 181–185.
- [95] T. K. Bhowmik, T. Paquet, and N. Ragot. 2014. OCR performance prediction using a bag of allographs and support vector regression. In *Proceedings of the International Workshop on Document Analysis Systems (DAS)*, pp. 202–206.
- [96] J. Kumar, P. Ye, and D. Doermann. 2013. A dataset for quality assessment of camera-captured document images. In *Proceedings of the International Workshop on Camera-Based Document Analysis and Recognition*, Springer, pp. 113–125.
- [97] X. Zhu and P. Milanfar. 2010. Automatic parameter selection for denoising algorithms using a no-reference measure of image content. *IEEE Trans. Image Process.*, 19, 12, pp. 3116–3132, doi: 10.1109/TIP.2010.2052820.
- [98] J. Shemiakina, E. Limonova, N. Skoryukina, V. V. Arlazarov, and D. P. Nikolaev. 2021. A method of image quality assessment for text recognition on camera-captured and projectively distorted documents. *Mathematics*, 9, 17, pp. 1–22, doi: 10.3390/math9172155.
- [99] X. Peng and C. Wang. 2020. Camera captured DIQA with linearity and monotonicity constraints, in *Proceedings of the International Workshop on Document Analysis Systems (DAS)*, X. Bai, D. Karatzas, and D. Lopresti, Eds., Springer International Publishing, pp. 168–181.
- [100] J. Xu, P. Ye, Q. Li, Y. Liu, and D. Doermann. 2016. No-reference document image quality assessment based on high order image statistics. In *Proceedings of the International Conference on Image Processing (ICIP)*, pp. 3289–3293.
- [101] H. Li, F. Zhu, and J. Qiu. 2019. Towards document image quality assessment: A text line based framework and a synthetic text line image dataset. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR19)*, pp. 551–558. doi: 10.1109/ICDAR.2019.00094.
- [102] P. Li, L. Peng, J. Cai, X. Ding, and S. Ge. 2017. Attention based RNN model for document image quality assessment. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR17)*, pp. 819–825.
- [103] R. Shinghal and C. Y. Suen. 1982. A method for selecting constrained hand-printed character shapes for machine recognition. *IEEE Trans. Pattern Anal. Mach. Intell.*, 4, 1, pp. 74–78, doi: 10.1109/tpami.1982.4767199. PMID: 21869007.
- [104] H. Hase, M. Yoneda, M. Sakai, and J. Yoshida. 1988. Evaluation of handprinting variation of characters using variation entropy. *IEICE Trans. D*, 71, 6, pp. 1048–1056.
- [105] H. Hase. 2011. Quality evaluation of character image database and its application. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR11)*, pp. 1414–1418.
- [106] Q. A. Bui, D. Molard, and S. Tabbone. 2018. Predicting mobile-captured document images sharpness quality. In *Proceedings of the International Workshop on Document Analysis Systems (DAS)*, pp. 275–280. doi: 10.1109/DAS.2018.32.
- [107] S. Xu, S. Jiang, and W. Min. 2017. No-reference/blind image quality assessment: A survey. *IETE Tech. Rev.*, 34, 3, pp. 223–245, doi: 10.1080/02564602.2016.1151385.
- [108] Z. Wang and A. C. Bovik, Eds. 2006. *Modern image quality assessment*. in *Synthesis Lectures on Image, Video, and Multimedia Processing (SLIVMP)*. Springer.
- [109] T. Tao, L. Ding, and H. Huang. 2020. Unified non-uniform scale adaptive sampling model for quality assessment of natural and screen images. *Neurocomputing*, 399, pp. 96–106.
- [110] Y. Fang, R. Du, Y. Zuo, W. Wen, and L. Li. 2020. Perceptual Quality Assessment for Screen Content Images by Spatial Continuity. *IEEE Trans. Circuits Syst. Video Technol.*, 30, 11, pp. 4050–4063, doi: 10.1109/TCSVT.2019.2951747.
- [111] K. Gu, J. Zhou, J.-F. Qiao, G. Zhai, W. Lin, and A. C. Bovik. 2017. No-Reference Quality Assessment of Screen Content Pictures. *IEEE Trans. Image Process.*, 26, 8, pp. 4005–4018, doi: 10.1109/TIP.2017.2711279.
- [112] P. Ye, J. Kumar, L. Kang, and D. Doermann. 2013. Real-time no-reference image quality assessment based on filter learning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 987–994. doi: 10.1109/CVPR.2013.132.
- [113] D. Schulz, J. Maureira, J. Tapia, and C. Busch. 2022. Identity Documents Image Quality Assessment. In *Proceedings of the 30th European Signal Processing Conference (EUSIPCO)*, pp. 1017–1021. doi: 10.23919/EUSIPCO55093.2022.9909773.
- [114] S. González and J. Tapia. 2022. Towards Refining ID Cards Presentation Attack Detection Systems using Face Quality Index. In *Proceedings of the 30th European Signal Processing Conference (EUSIPCO)*, pp. 1027–1031. doi: 10.23919/EUSIPCO55093.2022.9909529.
- [115] Z. Li, C. Yang, Q. Shen, and S. Wen. 1828. A document image dataset for quality assessment. *J. Phys. Conf. Ser.*, 1, p. 012033, 2021, doi: 10.1088/1742-6596/1828/1/012033.
- [116] A. Antonacopoulos, C. Clausner, C. Papadopoulos, and S. Pletschacher. 2011. Historical Document Layout Analysis Competition. In *Proceedings of the 2011 International Conference on Document Analysis and Recognition*, pp. 1516–1520.
- [117] N. Nayef, M. M. Luqman, S. Prum, S. Eskenazi, J. Chazalon, and J.-M. Ogier. 2015. SmartDoc-QA: A dataset for quality assessment of smartphone captured document images-single and multiple distortions. In *Proceedings of the International Conference on Document Analysis and Recognition (ICDAR15)*, pp. 1231–1235.
- [118] H. R. Sheikh, Z. Wang, L. K. Cormack, and A. C. Bovik. 2004. LIVE image quality assessment database release 2. [Online]. Available: <http://live.ece.utexas.edu/research/quality>.
- [119] E. C. Larson and D. M. Chandler. 2010. Most apparent distortion: full-reference image quality assessment and the role of strategy. *J. Electron. Imaging*, 19, 1, p. 011006.
- [120] N. Ponomarenko et al.. 2009. TID2008 - A database for evaluation of full-reference visual quality assessment metrics. *Adv. Mod. Radioelectron.*, 10, pp. 30–45.
- [121] A. Alaei, R. Raveaux, and D. Conte. 2017. Image quality assessment based on regions of interest. *Signal Image Video Process.*, 11, 4, pp. 673–680, doi: 10.1007/s11760-016-1009-z.
- [122] M. Darwiche, T.-A. Pham, and M. Delalandre. 2015. Comparison of JPEG's competitors for document images. In *Proceedings of the 2015 International Conference on Image Processing Theory, Tools and Applications (IPTA)*, pp. 487–493. doi: 10.1109/IPTA.2015.7367194.